

Drivers of interest rate dispersion in credit markets

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Credit markets exhibit substantial interest rate variation across firms and banks.

Most is explained by observables at loan issuance:

- for example: macroeconomic conditions, firm and contract characteristics and bank-firm relationships
- focus on explained component [Altavilla et al. (2024)]

However, a non-negligible share of variation remains unexplained

Motivation

The current literature is focusing on:

- dispersion **between similar firms and products**
- the role of **market inefficiencies**

Main argument: dispersion persists after controlling for credit risk → market inefficiencies

→ Search frictions, market power [Bocola et al., 2026 WP],[Amiti et al., 2026 WP]

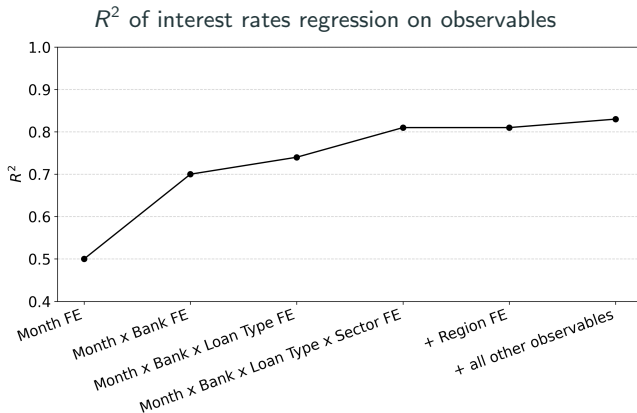
→ Firms' search abilities or randomness [Ferreira et al., 2024 WP]

Contributions

1. I add a **within-bank** dimension
 - dispersion as optimal pricing strategy of the bank
2. I control for **realized credit risk outcomes** of firms
 - unobserved risk captured by bank screening is a potential driver of residual dispersion
3. I consider **jointly** imperfect competition and bank screening
 - important implications for **surplus sharing** between banks and firms and **screening incentives**

Overview: Interest rate dispersion

1. Dispersion within banks, between similar firms and products.



- +15% remains unexplained
- Dispersion is large and time varying (approx. 1pp)

Overview: Empirical approach

2. I use realized firms' outcomes to control for unobserved risk

1. Unexplained rate residuals **positively correlate with negative surprises in firms' future credit risk**

- stronger when information is more valuable (during crisis, for young, small firms)

2. Test for a potential **leverage effect**

- test the effect of a rise in rates on firms' riskiness exploiting fixed-variable rate structures
- find no evidence that firms' increased debt-servicing costs can trigger a deterioration in their creditworthiness

→ **suggests that the correlation is driven by banks' active screening**

3. I consider jointly imperfect competition and bank screening

Two banks, two firm types with different riskiness. Dispersion is *jointly* determined by:

- **Imperfect competition** (search frictions) → *within-type* dispersion
- **Costly signal acquisition** → *between-type* dispersion

Implications on surplus sharing:

- Adverse selection is costly for banks, and for safer firms (cross-subsidy, exit), while it benefits riskier firms.
- Banks' share of total surplus is decreasing in competition and increasing in the precision of the signal
- ↑ information reduces (increases) costs borne by safer (riskier) firms

Method: SMM. Pin down the relative contribution of search frictions and information to residual dispersion

Findings:

1. Information explains little of the residual dispersion in the French market
 - → search frictions account for $> 95\%$ of residual rate variance
 - → near-perfect screening in safe rating classes, none in risky ones
2. but determines heterogeneous allocation of surplus across firm types and banks
 - → riskier types gain the most when adverse selection is mild and banks do not screen.

Empirical Strategy

Firm-level datasets

1. Loan-level survey (NCE, 2006–2024)

Loan-level survey: all new loan issuance on the 1st month of each quarter

Includes banks' id, loan terms and firms' accounting, credit, and risk data

Cannot observe: full loan history for firms.

2. Credit Registry

Tracks outstanding stock of credit across banks-firms (thresholds: loan

exposures $\geq$ €25,000 and/or firm balance-sheet $\geq$ €750,000)

3. Balance Sheets (FIBEN)

Annual accounting data from tax filings. Firms' demographics, additional controls.

4. Banque de France credit ratings (COTA)

Forward-looking over 3 years; incorporates quantitative and qualitative data, historical trends and payment incidents

I turn ratings into default probabilities using historical default frequencies



Realized credit risk outcome

I use unexpected changes in default risk as a proxy for true type.

I estimate

$$r_{ilbt} = \alpha + \sum_{k=1}^K \beta_k \Delta P_{i,t+k} + \mathbf{x}'_{ilbt} \gamma + \mu_{bt\tau s} + \varepsilon_{ilbt}, \quad (1)$$

r_{ilbt} is the rate on loan l granted by bank b to firm i in quarter t

$\Delta P_{i,t+k}$ is the change in default probability of firm i between $k - 1$ and k years after issuance.

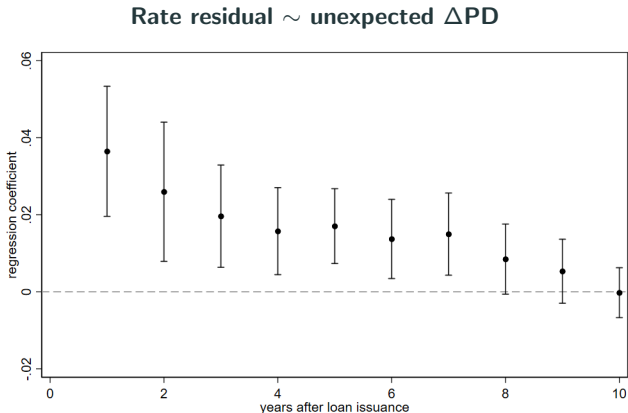
$\mathbf{x}_{ilbt}$ holds what the bank observes at issuance

$\mu_{bt\tau s}$ absorbs month $\times$ lender $\times$ loan-type $\times$ two-digit-sector effect

Balanced sample, non-defaulting firms at time t

►► actual default

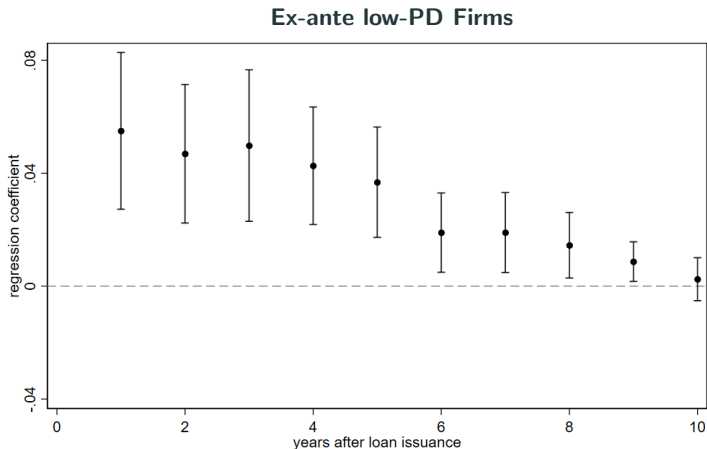
Correlation residuals vs default probability



β_k at horizon k , with 95% CI. All $K = 10$ surprises enter jointly; balanced sample.

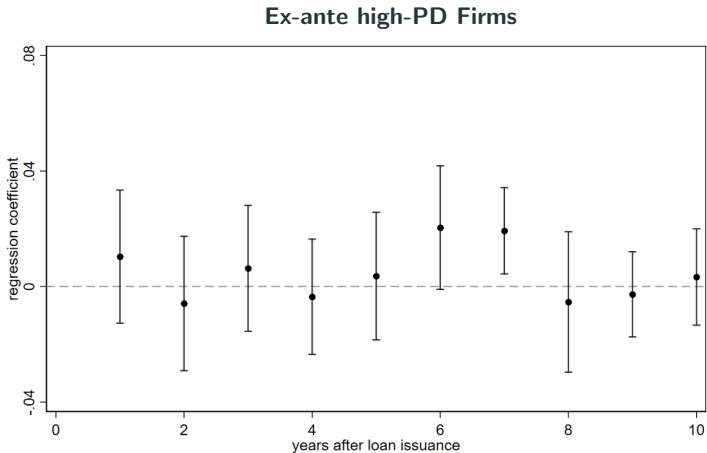
- Positive and significant, weaker at longer horizons

Correlation: stronger for better rated firms



Computed separately for firms with ex-ante $PD < 0.62\%$ vs $PD > 3.45\%$.

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Information, or a leverage effect?

A positive correlation admits two readings:

Information — the bank recognizes the riskier type and prices it

Leverage — the high rate itself worsens the firm's fundamentals

Information, or a leverage effect?

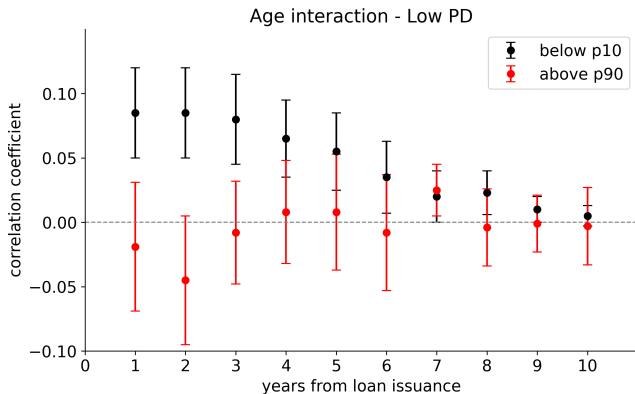
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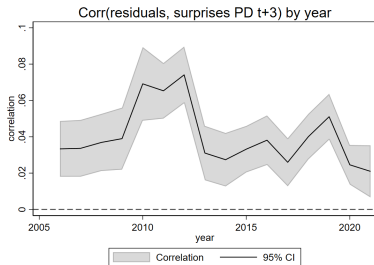
Correlation: stronger for younger firms

Focusing on the group of better rated firms (low PD):

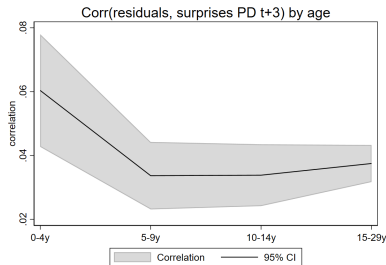


Excludes firms in liquidation. Balanced sample. Firms with ex-ante good rating. Interactions at 10th-90th percentile of age.

Correlation: demographics



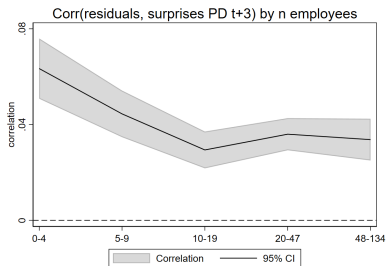
(a) By year



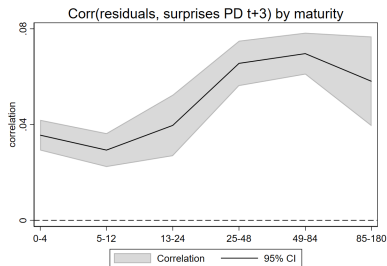
(b) By firm age

Figure 1: Correlation $\hat{\rho}_g$ between the unexplained rate component and the unexpected 3-year change in default probability (ΔPD_{t+3}), both computed in the entire population and correlated by group g .

Correlation: demographics



(a) By n. of employees



(b) By loan maturity (months)

Figure 2: Correlation $\hat{\rho}_g$ between the unexplained rate component and the unexpected 3-year change in default probability (ΔPD_{t+3}), both computed in the entire population and correlated by group g .

→ the informational content is strongest where adverse selection is worst.

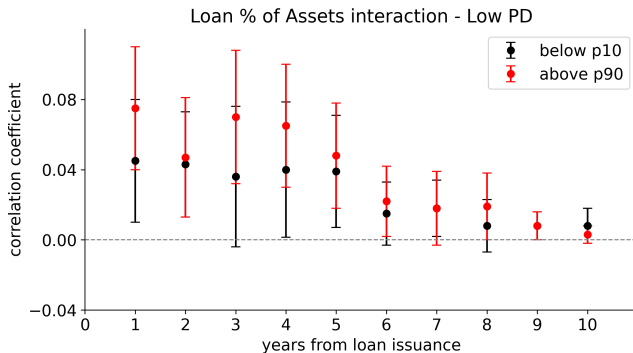
Information, or a leverage effect?

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- Leverage** — the high rate itself worsens the firm's fundamentals

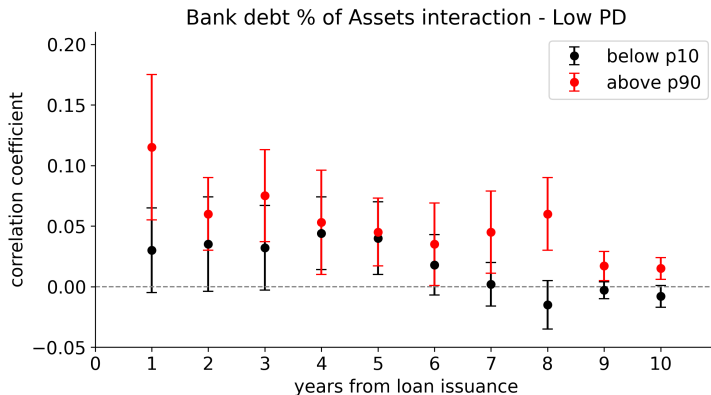
Correlation: does not depend on loan size

Focusing on the group of better rated firms (low PD):



Excludes firms in liquidation. Balanced sample. Firms with ex-ante good rating. Interactions at 10th-90th percentile of loan over assets.

Higher PD. Prediction or consequence?



Probability of default, by year. Excludes firms in liquidation. Balanced sample. Rating: 1 to 3, from best to worst rating. Interactions at 10th-90th percentile of bank debt over assets.

Test: do higher rates induce rating worsening?

Design. Compare **floating**- and **fixed**-rate borrowers when reference rates rise.

Assumption. The increase is set by aggregate conditions and monetary policy, so it is exogenous to the individual firm's fundamentals.

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$$\begin{aligned}\Delta PD_{i,t+3} = & \beta \Delta \text{Ref}_{l,t \rightarrow t+3} + \delta (\Delta \text{Ref}_{l,t \rightarrow t+3} \times \text{Share}_{i,t}) \\ & + \mathbf{x}'_{ilbt} \gamma + \mu_{bt\tau s} + \varepsilon_{ilbt}\end{aligned}$$

$\Delta PD_{i,t+3} = 100 \times (P_{i,t+3} - P_{i,t})$; $\text{Share}_{i,t}$ = loan- or bank-debt-over-assets.

$\Delta \text{Ref}_{l,t \rightarrow t+3}$ is loan l 's **own** reference rate change: each floating loan is tied to its own index (EONIA, Euribor 1m/3m/1y, TMO), a fixed-rate loan to zero.

Maturity ≥ 30 months; SEs two-way clustered by lender and month.

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Maturity ≥ 30 months; SEs two-way clustered by lender and month.

Variation within a lender $\times$ month $\times$ loan-type $\times$ sector cell: fixed vs floating, and *which index* a floating loan carries.

Test: do higher rates induce rating worsening?

	Default Probability						
Variable rate (dummy)	0.1176** (0.014)		0.1407 (0.127)	0.1039** (0.026)			
cum dRate end y2		0.0433 (0.229)			0.0374 (0.254)		
annual dRate end y1			0.0812 (0.600)			0.0524 (0.711)	0.0425 (0.773)
annual dRate end y2			0.0567 (0.330)			0.0363 (0.551)	0.0128 (0.879)
Variable rate (dummy) × Loan over assets				0.1932 (0.370)			
cum dRate end y2 × Loan over assets					0.0711 (0.264)		
annual dRate end y1 × Loan over assets						-0.1330 (0.763)	
annual dRate end y2 × Loan over assets						0.0905 (0.148)	
annual dRate end y1 × Bank Debt share							0.0058 (0.980)
annual dRate end y2 × Bank Debt share							0.0986 (0.623)
Loan over assets	0.2250** (0.030)	0.2002** (0.035)	0.2048** (0.035)	0.1808* (0.094)	0.1894* (0.054)	0.1849* (0.059)	0.1929* (0.054)
Bank Debt share	0.0846** (0.040)	0.1139** (0.019)	0.1134** (0.019)	0.0850** (0.034)	0.1118** (0.022)	0.1114** (0.023)	0.1127** (0.025)
Lender × Month × Loan Type × 2digit Sector FE	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓
All other observables	✓	✓	✓	✓	✓	✓	✓
Observations	102,463	96,414	96,414	102,463	96,414	96,414	96,414
R ²	0.42	0.40	0.40	0.42	0.40	0.40	0.40

p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

Table 1: Data: Nouveaux Crédits (BdF), 2006q1-2024q2. Term loans, at least 30-month maturity, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

Test: do higher rates induce rating worsening? (2021q3–q4)

	Default Probability - 2021q3-2021q4 only						
Variable rate (dummy)	-0.3953 (.)		-38.7291 (0.419)	-0.3499 (.)			
annual dRate end y2			9.8382 (0.420)		0.1178 (0.929)	1.0014 (0.249)	
annual dRate end y3			6.3408 (0.456)		-0.5977 (0.720)	-1.9467 (0.156)	
Variable rate (dummy) × Loan over assets				-0.5306 (.)			
annual dRate end y2 × Loan over assets					6.1711 (0.743)		
annual dRate end y2 × Bank Debt share						-1.2526 (0.215)	
annual dRate end y3 × Loan over assets					-9.1385 (0.732)		
annual dRate end y3 × Bank Debt share						1.8861 (0.214)	
Loan over assets	-0.7120 (.)	-1.4514 (.)	-1.4851 (0.236)	-0.3328 (.)	-0.9459 (.)	-0.9777 (0.400)	-1.5080 (0.222)
Bank Debt share	0.1002 (.)	-0.2848 (.)	-0.2949 (0.439)	0.1286 (.)	-0.2557 (.)	-0.2801 (0.456)	-0.3168 (0.585)
Lender × Month × Loan Type × 2digit Sector FE	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓
All other observables	✓	✓	✓	✓	✓	✓	✓
Observations	4,112	3,895	3,895	4,112	3,895	3,895	3,895
R ²	0.42	0.42	0.42	0.42	0.42	0.42	0.42

p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

Table 2: Data: Nouveaux Crédits (BdF), 2006q1-2024q2. Term loans, maturity at least 30 months, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

Rate exercise: 2021q3–q4, 2022 increase only

	Default Probability - 2021q3-2021q4 only						
Variable rate (dummy)	-0.5698 (0.169)	-6.1935 (0.193)	-0.5121 (0.186)				
annual dRate end y2		1.8860 (0.261)			-0.3828* (0.095)	-0.4445 (0.218)	
Variable rate (dummy) × Loan over assets			-0.6271 (0.249)				
annual dRate end y2 × Loan over assets					-0.3410 (0.228)		
annual dRate end y2 × Bank Debt share							0.0555 (0.915)
Loan over assets	-0.5329 (0.374)	-1.1142 (0.168)	-1.1243 (0.216)	-0.1584 (0.740)	-0.6316 (0.224)	-0.6318 (0.223)	-1.1395 (0.177)
Bank Debt share	0.0986 (0.725)	-0.2614 (0.316)	-0.2672 (0.448)	0.1401 (0.631)	-0.2234 (0.375)	-0.2234 (0.375)	-0.2780 (0.364)
Lender × Month × Loan Type × 2digit Sector FE	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓
All other observables	✓	✓	✓	✓	✓	✓	✓
Observations	4,324	4,089	4,089	4,324	4,089	4,089	4,089
R ²	0.41	0.41	0.41	0.41	0.41	0.41	0.41

p-values in parentheses
 * p< 0.10, ** p< 0.05, *** p< 0.01

Table 3: Data: Nouveaux Crédits (BdF), 2006q1-2024q2. Term loans, maturity at least 24 months, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

►► 2022–23 increase, mat. ≥ 30m

►► 2022–23 increase, mat. ≥ 24m

Test: do higher rates induce rating worsening?

Result: β and δ are not significant.

→ higher debt-servicing costs do **not** make firms riskier

→ the loan rate nonetheless **remains** a significant predictor of ex-post risk

Theory

Environment

Market with search frictions and adverse selection as in [Lester et al. 2019]

- small I depart from screening via menus and focus on information acquisition.

A continuum of firms: visit banks to obtain rate quotes

- **asymmetric information:** two types $i \in \{g, b\}$, measure μ_q^i , default with probability $1 - p^i$ accept maximum rate $\bar{r}^i$
- **search frictions:** π shoppers ($1 - \pi$ captive) firms obtain two (one) rate quotes, and accept the lowest one, if lower than $\bar{r}^i$

Two homogeneous banks:

- cost of lending depends on central bank rate and risk: $c^i = \frac{r^{CB}}{p^i}$
- cannot distinguish captive/shopper firms, nor types
- can choose to acquire a **costly, symmetric signal** s on all firms, with precision $q \in [0.5, 1]$

Symmetric equilibrium CDF of rates $F^{s_i}(r)$

After acquiring signal s_i of precision q , the bank forms an expectation of:

- **posterior of good type share** $\alpha^{s_i} = \Pr(\mathbf{g} \mid \mathbf{s}_i)$

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Bank chooses CDF of rates $F^{s_i}(r)$ to maximize expected profits:

$$\Pi^{s_i}(r) = \begin{cases} \underbrace{[1 - \pi + \pi(1 - F^{s_i}(r))]}_{\text{acceptance probability}} \underbrace{(r - c^{avg, s_i})}_{\text{profit per trade}} & \text{if } r \leq \bar{r}^g \\ 0 & \text{if } r > \bar{r}^g \end{cases}$$

- below $\bar{r}^g$, **both types accept**
- **Trade-off:** a higher rate raises the profit per trade and lowers the probability of acceptance

Symmetric equilibrium CDF of rates $F^{s_i}(r)$

After acquiring signal s_i of precision q , the bank forms an expectation of:

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- **between $\bar{r}^g$ and $\bar{r}^b$, only bad types accept**
- **Trade-off:** $\uparrow \text{rate} \rightarrow \uparrow \text{profit per trade and } \downarrow \text{probability of acceptance}$

Where does the bank lend?

It depends on the severity of adverse selection ϕ .

$$\phi \equiv (\bar{r}^g - c^{avg}) - \mu_b(\bar{r}^b - c_b)$$

Intuition: compares profits in pooling vs profits when lending to bad types only

$$\begin{cases} \phi > 0 & \text{mild: the bank } \mathbf{pools} \text{ both types} \\ \underline{\phi} < \phi \leq 0 & \text{severe: a } \mathbf{mix} \text{ of pooling and bad-only} \\ \phi \leq \underline{\phi} & \text{severe: } \mathbf{bad \ types \ only} \end{cases}$$

with $\underline{\phi} = -\frac{\pi}{1-\pi}(\bar{r}^g - c^{avg}) < 0$, found from $F^a(\bar{r}^g) > 0$.

Rate dispersion

Limit cases of $\pi = 0, 1$ ¹

- Banks operate under monopoly/perfect competition
- degenerate distribution

► Adverse selection

¹ π is the share of shoppers

Rate dispersion

Limit cases of $\pi = 0, 1$ ¹

- Banks operate under monopoly/perfect competition

→ degenerate distribution

Intermediate case $\pi \in (0, 1)$

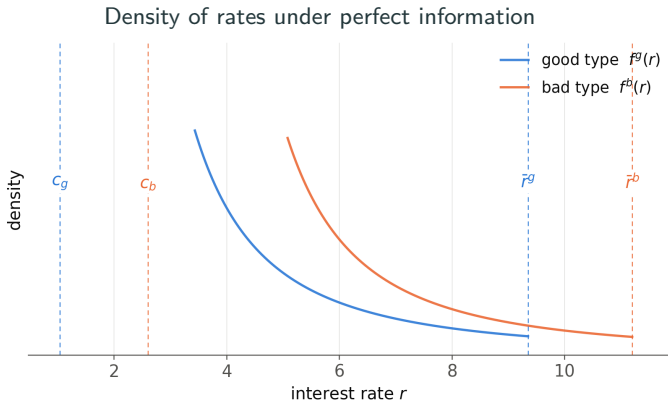
- at any r^* , a bank deviating to $r^* - \varepsilon$ gains all shoppers → Strict gain
- at $r^* = c$, $\Pi^* = 0$. A bank deviating to any $r > c$ makes profits from captive firms

→ dispersed equilibrium

» Adverse selection

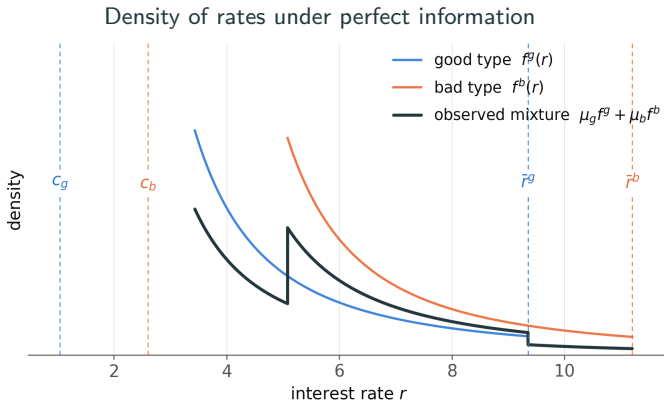
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Example: two types, perfect signal ($q = 1$)



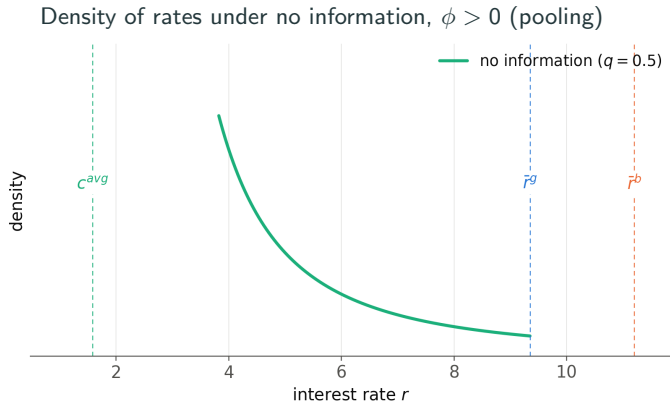
- risky types cost more (c_b) and tolerate more ($\bar{r}^b$)
- With $q = 1$ the bank solves **two separate problems**, one per type

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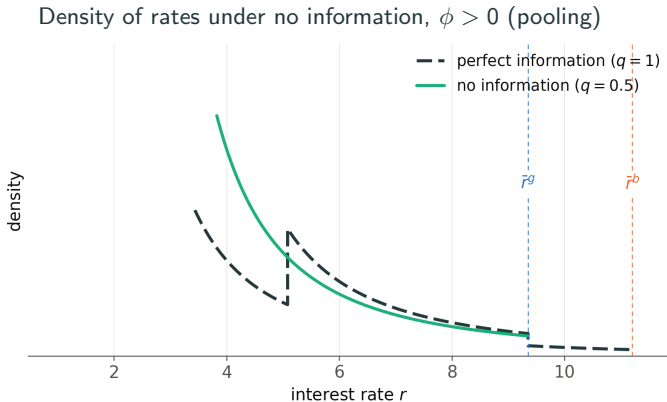


- risky types cost more (c_b) and tolerate more ($\bar{r}^b$)
- With $q = 1$ the bank solves **two separate problems**, one per type
- observed rates are the mixture $\mu_g f^g(r) + \mu_b f^b(r)$

Example: two types, no signal acquisition ($q = 0.5$)



Example: two types, no signal acquisition ($q = 0.5$)



- Possible outcomes: pooling, bad-only lending, or a mix depends on the degree of adverse selection ϕ
- both types face the **same** rates $\rightarrow$ no between-type dispersion

Back to the data: variance and correlation

Dispersion

$$q = 1 \quad \text{Var}(r) = \underbrace{\sum_{i \in \{g, b\}} \mu_i \text{Var}(r | i)}_{\text{within-type}} + \underbrace{\mu_g \mu_b (\mathbb{E}[r|b] - \mathbb{E}[r|g])^2}_{\text{between-type}}$$

$$q = 0.5, \phi > 0 \quad \text{Var}(r)_{q=0.5} < \text{Var}_{\text{within}} < \text{Var}(r)_{q=1}$$

Back to the data: variance and correlation

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$$q = 0.5, \phi > 0 \quad \text{Var}(r)_{q=0.5} < \text{Var}_{\text{within}} < \text{Var}(r)_{q=1}$$

Correlation rates and riskiness

$$q = 1 \quad \rho(r, PD) = \sqrt{\mu_g \mu_b} \frac{\mathbb{E}[r|b] - \mathbb{E}[r|g]}{\sqrt{\text{Var}(r)}}$$

$$q = 0.5, \phi > 0 \quad 0$$

Back to the data: variance and correlation

Dispersion

$$q = 1 \quad \text{Var}(r) = \underbrace{\sum_{i \in \{g, b\}} \mu_i \text{Var}(r | i)}_{\text{within-type}} + \underbrace{\mu_g \mu_b (\mathbb{E}[r|b] - \mathbb{E}[r|g])^2}_{\text{between-type}}$$

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Correlation rates and riskiness

$$q = 1 \quad \rho(r, PD) = \sqrt{\mu_g \mu_b} \frac{\mathbb{E}[r|b] - \mathbb{E}[r|g]}{\sqrt{\text{Var}(r)}}$$

$$q = 0.5, \phi > 0 \quad 0$$

→ I will target **Var** and **Corr** to calibrate π and q

Who gets the surplus?

Proposition 1 (Banks' share of total surplus)

The banks' share of total surplus is decreasing in competition and increasing in the precision of the signal.

Who gets the surplus?

Proposition 1 (Banks' share of total surplus)

The banks' share of total surplus is decreasing in competition and increasing in the precision of the signal.

Under a perfect signal every firm trades, each type is priced off its own cost, and the bank captures exactly $1 - \pi$ of the realized surplus, firms π . Under no information the bank collects strictly less:

$$\mathbb{E}[\Pi(q)] \propto (1 - \pi), \quad \Delta\Pi = \begin{cases} (1 - \pi) \mu_b (\bar{r}^b - \bar{r}^g) & \text{if } \phi > 0 \\ (1 - \pi) \mu_g (\bar{r}^g - c_g) & \text{if } \phi \leq 0 \end{cases}$$

$\phi > 0$: the bank pools and cannot extract from bad types. $\phi \leq 0$: good types exit.

...and what do firms get?

Proposition 2 (Firms' surplus and information)

Information reduces (increases) the costs borne by good (bad) types.

- (i) *Under no information, good types trade at the higher **pooling** rates when adverse selection is mild ($\phi > 0$), and **may not trade at all** when it is severe ($\phi \leq 0$). Their gains rise with q and fall with competition.*
- (ii) *Bad types instead **benefit** from pooling when $\phi > 0$, trading below $\bar{r}^g$; for $\underline{\phi} < \phi \leq 0$ they gain partly from cross-subsidization, less than under pooling; they gain nothing once $\phi \leq \underline{\phi}$ and the bank lends to bad types only. Their benefits fall with q .*

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Good types bear two costs of asymmetric information — a **cross-subsidy** to bad types where the bank pools, and **exclusion** where it lends to bad types only. Both fall with q .

When does a bank pay to screen?

Proposition 3 (Signal acquisition)

A bank acquires precision q if and only if

$$\mathbb{E}[\Pi(q)] - \mathbb{E}[\Pi(q = 0.5)] \geq C_s(q).$$

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$$\mu_b(1 - \pi)(\bar{r}^b - \bar{r}^g) \geq C_s(1) \quad \text{if } \phi > 0$$

$$\mu_g(1 - \pi)(\bar{r}^g - c_g(\bar{r}^g)) \geq C_s(1) \quad \text{if } \phi \leq 0$$

— the surplus it fails to extract from *bad* types, or the surplus lost to *good* types who exit.

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Corollary 1 (Competition and the return to screening)

$\mathbb{E}[\Pi(q)] \propto (1 - \pi)$ at every q , so the return to screening falls one for one with competition and vanishes as $\pi \rightarrow 1$: under perfect competition no bank pays a positive $C_s(q)$ for any precision.

Calibration

Calibration summary

Approach

- calibrated directly from the data: μ_i , PD_i , $\bar{r}^b$
- estimated by SMM within each rating class: π , q , $\bar{r}^g$
- moments matched: mean, variance, correlation with PD of rate residuals

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Results

- banks acquire a **very precise signal**, but its contribution to total unexplained dispersion is **at most 5%**
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Robustness

- local identification checked; competing global optima ruled out

» Approach

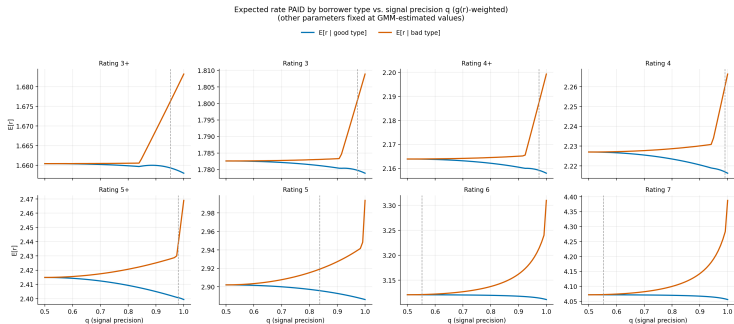
» Results

» Parameters

Surplus sharing

Who pays what, as screening improves

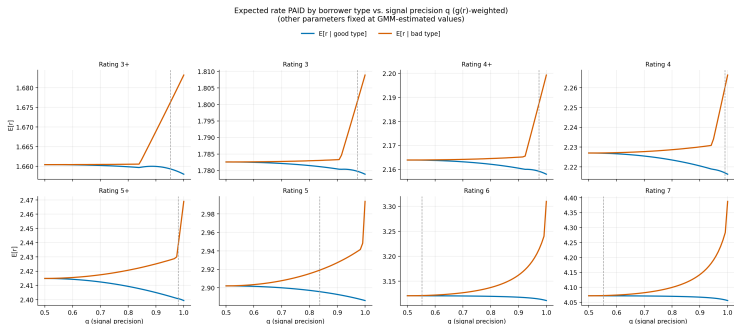
I hold $\pi, \bar{r}^g, \bar{r}^b, \mu_b, p_g, p_b$ at their calibrated values and vary q over $[0.5, 1]$.



Expected rate paid by good vs. bad types, by rating class, under the realized (transaction-weighted) distribution $g(r)$. Dashed line: the class's own $\hat{q}$.

Who pays what, as screening improves

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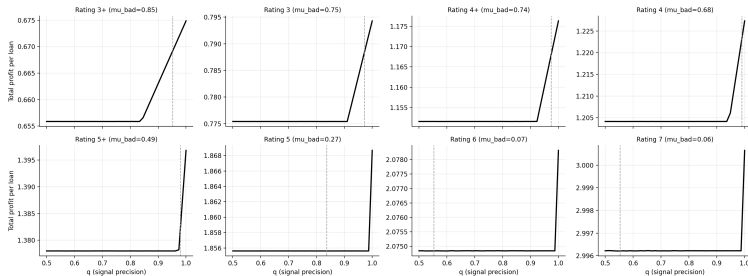


Expected rate paid by good vs. bad types, by rating class, under the realized (transaction-weighted) distribution $g(r)$. Dashed line: the class's own $\hat{q}$.

- at $q = 0.5$ every class **pools**: both types pay the same rate
- the gap opens **late**, once ϕ^{Si} turns negative and pooling is dropped

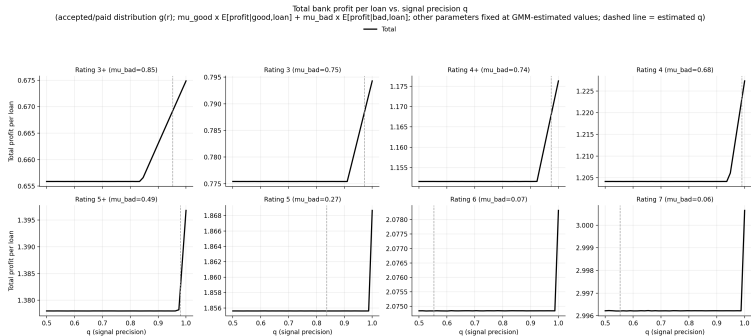
The bank's return to screening

Total bank profit per loan vs. signal precision q
 (accepted/paid distribution $g(r)$; $\mu_{\text{good}} \times E[\text{profit}|\text{good,loan}] + \mu_{\text{bad}} \times E[\text{profit}|\text{bad,loan}]$; other parameters fixed at GMM-estimated values; dashed line = estimated q)
 — Total



Total expected bank profit per loan, $\mu_g E[\text{profit} | g] + \mu_b E[\text{profit} | b]$, by rating class. Dashed line: the class's own $\hat{q}$.

The bank's return to screening



Total expected bank profit per loan, $\mu_g E[\text{profit} | g] + \mu_b E[\text{profit} | b]$, by rating class. Dashed line: the class's own $\hat{q}$.

- profit is **flat while the bank pools** — precision it cannot act on is worth nothing
- it rises only once the bank **separates types**, and peaks at $q = 1$
- at $q = 1$ the type gap reaches 0.025 (rating 3+), 0.20 (rating 6), 0.33 (rating 7)

Who gains from a perfect signal?

Change in surplus from $q = 0.5$ to $q = 1$, % of the class's total surplus

Rating	Bad types	Banks	Good types
3+	-0.892	0.881	0.010
3	-0.847	0.833	0.013
4+	-0.806	0.786	0.020
4	-0.958	0.907	0.054
5+	-0.646	0.550	0.095
5	-0.438	0.274	0.163
6	-0.217	0.069	0.148
7	-0.308	0.071	0.236

Rows sum to zero: every class is in full pooling at $q = 0.5$, so no firm is rationed at either end and information only redistributes.

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- information moves surplus from **bad types to banks**
- good types gain in every class, but least where screening is already near-perfect

Conclusion

Main Takeaways

Appendix

Example Documents Required for the Application

- Business plan including financial projections (forecast income statement, working capital requirements, financing plan)
- Company information: legal status, share capital, partners/shareholders
- Proof of personal contribution or equity investment
- Proof of guarantees or collateral (optional for “solidarity” loans)
- Documentation of the project cost (quotes, sales offers, franchise agreements)
- Company registration extract (Kbis) if the company is already registered

Survey Evidence: Banks

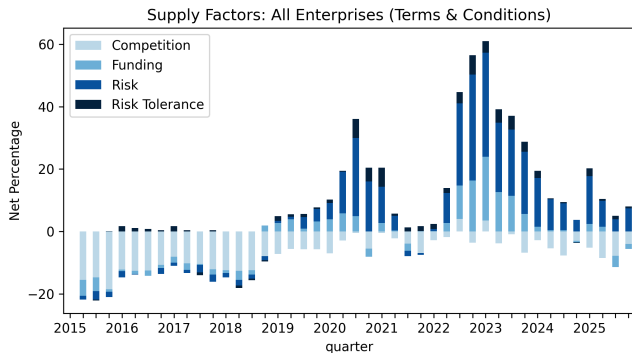


Figure 3: Source: BLS (ECB). Decomposition of Supply-Side Drivers for Loan Terms and Conditions tightening/easing, 3-month backward-looking, country-weighted. Net percentage contribution of primary supply-side factors to the tightening or easing of credit terms and conditions for new loans to enterprises in the Euro Area.

SAFE (ECB):

- 15–30% of firms applying for loans report an increase in financing costs (loan covenants, required guarantees, information requirements, procedures, time required for loan approval)
- 5–14% of firms report financing obstacles to obtaining a bank loan (rejections, high cost, limited amount granted, not applying for fear of rejection)

Baseline Specification

Table 4: Data: Nouveaux Credits (BdF), 2006q1-2024q2. Excludes credit lines and undefined maturity products. All credit and balance-sheet variables are expressed in terms of reported assets.

	Total effective rate					
log.Loan quantity (K-Eur)	-0.2219*** (0.000)	-0.1535*** (0.000)	-0.1537*** (0.000)	-0.1436*** (0.000)	-0.1467*** (0.000)	-0.0989** (0.000)
Maturity (months)	-0.0005* (0.073)	0.0022*** (0.002)	0.0032*** (0.001)	0.0021* (0.003)	0.0022*** (0.003)	0.0020*** (0.004)
Age	-0.0141*** (0.000)	-0.0095*** (0.000)	-0.0084*** (0.000)	-0.0074*** (0.000)	-0.0073*** (0.000)	-0.0038*** (0.000)
Age squared	0.0001*** (0.000)	0.0000*** (0.000)	0.0000*** (0.000)	0.0000*** (0.000)	0.0000*** (0.000)	0.0000*** (0.000)
Young firm 0-3y (dummy)	-0.0577*** (0.001)	-0.0404 (0.297)	-0.0424 (0.285)	-0.0190 (0.528)	-0.0161 (0.575)	-0.0669** (0.012)
New bank-firm relationship (dummy)	0.1988*** (0.000)	0.0183 (0.639)	0.0145 (0.715)	-0.0171 (0.293)	-0.0179 (0.285)	-0.0403** (0.025)
Variable rate (dummy)	0.4528*** (0.000)	0.4905*** (0.000)	0.1665* (0.051)	0.1184 (0.196)	0.1167 (0.196)	0.1434 (0.119)
Personal loan (dummy)	-0.2999** (0.035)	-0.0612 (0.674)	0.0400 (0.620)	-0.0066 (0.943)	-0.0043 (0.962)	0.0603 (0.447)
Individual entrepreneur (dummy)	0.5724*** (0.000)	0.5680*** (0.000)	0.5333*** (0.000)	0.4320*** (0.000)	0.4195*** (0.000)	0.2404*** (0.001)
Loan over assets	0.1613** (0.021)	0.0983* (0.064)	0.0986* (0.080)	0.1057* (0.079)	0.1039* (0.076)	0.0474* (0.086)
Pledged guarantees	0.8325*** (0.003)	0.1146 (0.132)	-0.0204 (0.709)	-0.2563** (0.016)	-0.2360** (0.011)	-0.1759* (0.025)
Revenues	-0.0444*** (0.000)	-0.0165 (0.367)	-0.0114 (0.367)	0.0034 (0.592)	0.0050 (0.449)	-0.0073 (0.180)
Bank Debt	0.1594*** (0.000)	0.1219** (0.008)	0.1158** (0.014)	0.1097*** (0.001)	0.1218*** (0.001)	-0.0001 (0.994)
Non-Bank Debt	-0.0395 (0.401)	-0.0583 (0.630)	-0.0455 (0.723)	-0.0055 (0.945)	-0.0009 (0.991)	-0.0167 (0.742)
Mobilised credit	-0.0117* (0.023)	-0.0021 (0.471)	0.0013 (0.700)	0.0000 (0.991)	0.0003 (0.927)	0.0025 (0.300)
Short Term Debt	0.0009 (0.967)	-0.0024 (0.871)	-0.0009 (0.951)	0.0190 (0.159)	0.0190 (0.159)	-0.0125 (0.283)
Undrawn credit lines (K-Eur)	-0.0109 (0.967)	-0.0058 (0.871)	-0.0044 (0.951)	-0.0070 (0.159)	-0.0071 (0.159)	-0.0025 (0.283)
Quarter FE	✓					
Lender × Quarter FE		✓				
Lender × Quarter × Loan Type FE			✓			
Lender × Quarter × Loan Type × 2digit Sector FE				✓		
Region FE					✓	
Group indicator, Rating FE						✓
Observations	457,868	457,868	457,868	457,868	457,868	457,868
R ²	0.55	0.72	0.76	0.82	0.82	0.83

p-values in parentheses

* p < 0.10, ** p < 0.05, *** p < 0.01

The regression sample

Rating Scale	N obs.	rate (mean)	rate (s.d.)	Quantity (mean, K-Eur)	Maturity (mean, months)	Var. Rate (in %)	Age (mean, years)	Loan/Assets (mean, in %)	Bank Debt/Assets (mean, in %)
3++	7948	2.1	1.8	3754.5	59.0	35	39.0	5.4	19.6
3+	22681	2.2	1.8	2919.3	53.5	37	33.7	5.0	20.6
3	43073	2.3	1.8	2271.5	42.4	37	31.7	5.3	22.0
4+	71629	2.8	2.1	1092.2	34.8	39	28.1	6.2	23.0
4	112974	2.8	2.0	670.2	27.9	43	26.1	7.1	25.0
5+	106418	2.9	2.0	545.5	23.9	48	23.4	7.7	30.3
5	46510	3.5	2.2	781.8	17.9	51	23.1	8.3	31.0
6	13625	3.7	2.2	923.4	16.6	52	21.4	10.4	34.2
7	2657	4.2	2.3	134.7	14.8	62	19.7	8.6	32.7
8	1592	5.0	2.6	107.8	13.2	61	18.0	8.4	31.2
9	268	5.5	2.7	134.0	11.9	48	16.5	10.0	31.7
P (liquidation)	3716	5.0	2.0	114.5	3.4	26	21.8	5.7	33.4
0 (no rating)	24777	3.2	2.1	1895.8	45.0	40	16.1	12.4	27.2
Total	457868	2.9	2.1	1098.1	30.5	44	25.8	7.3	26.5

Table 5: Descriptive table of the loan-level regression sample. Source: Banque de France (Nouveaux Credits, COTA, SCR, FIBEN). Rate: total effective rate (including all fees).

458 thousand loans, 120 thousand firms, 2006q1–2022q4. Fixed-term products to private companies; credit lines and undefined-maturity credit excluded.

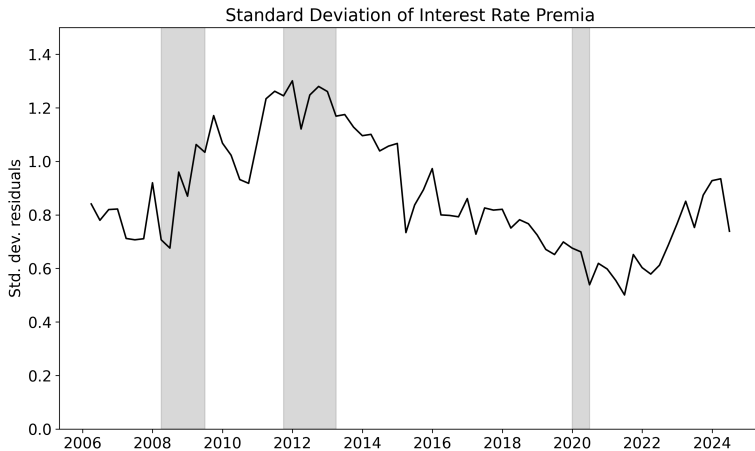
Measuring dispersion: empirical approach

Extract the **unexplained component of rates** by controlling for:

- **Interacted Fixed Effects** $\delta_{b,p,s,t}$: Interact month (t), bank (b), loan type (p), sector (s)
- **Loan (l) characteristic**: amount, maturity, loan purpose, variable rate, personal use, guarantees pledged, bundles
- **Firm (f) characteristic**: region/department, age, new bank relationship, sector, credit rating, group indicator, individual entrepreneur, credit&accounting data

$$\text{Rate}_{f,b,l,t} = \alpha + \beta \cdot \text{LoanChar}_{f,b,l,t} + \gamma \cdot \text{FirmChar}_{f,t} + \delta_{b,p,s,t} + \varepsilon_{f,b,l,t}$$

Rate residuals distribution: standard deviation



Source: Nouveaux Crédits, BdF, 2006q1-2024q2.

Baseline specification

How are observable characteristics correlated with the rate?

1. **Negatively correlated:** Loan quantity, if Startup, Age, New Relationship reported in Credit Registry, Pledged guarantees, Fraction of drawn credit over assets
2. **Positively correlated:** Maturity, Individual entrepreneur, Bank risk weighting
3. **Not significantly correlated:** Variable or fixed rate, if Personal Loan, Loan over assets, Revenues, Total Bank debt over assets, Short-term debt over assets, Undrawn credit over assets

Baseline Specification

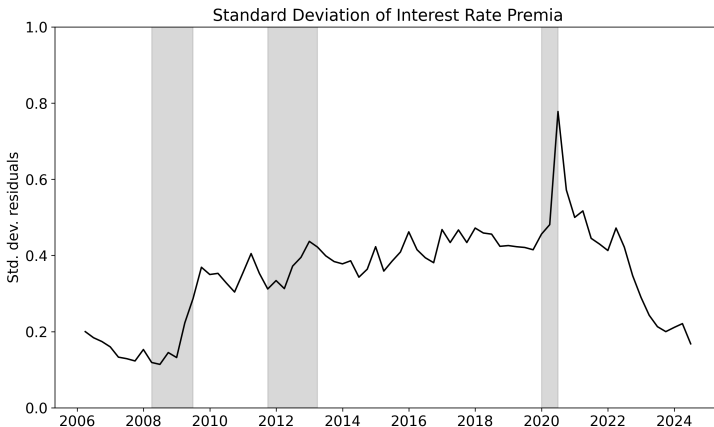
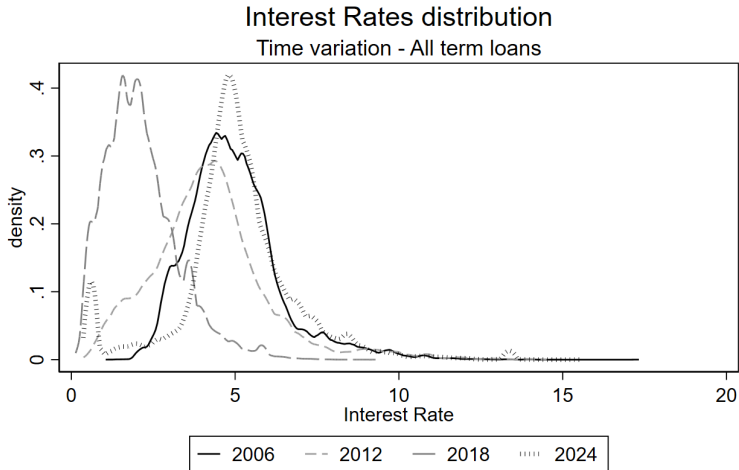


Figure 4: Source: Nouveaux Crédits, BdF, 2006q1-2024q2.

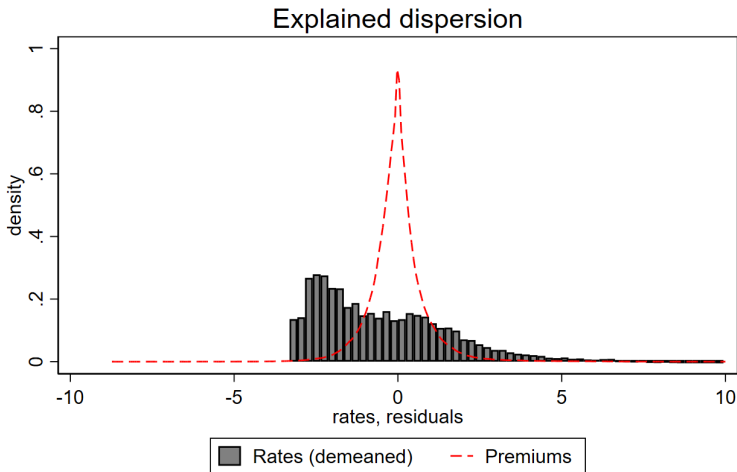
1% s.d. has a different meaning if rates are at 2% vs 8% → I compute the s.d. of the residuals over the mean rate.

Rates dispersion



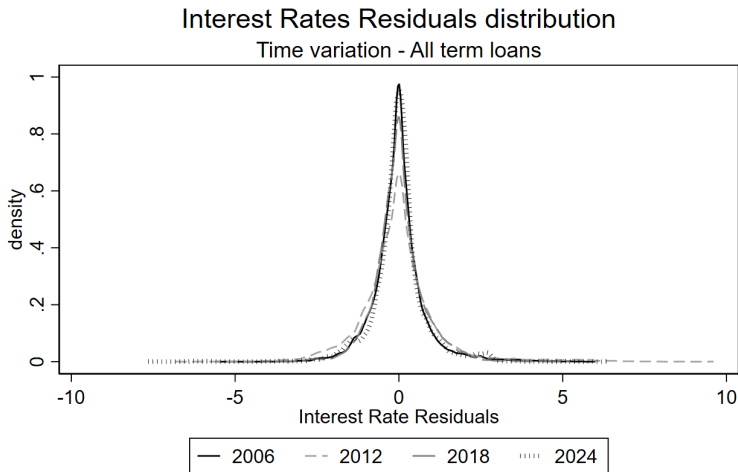
Source: Nouveaux Credits, BdF, 2006q1-2024q2
Excludes credit lines and personal loans.

Explained dispersion



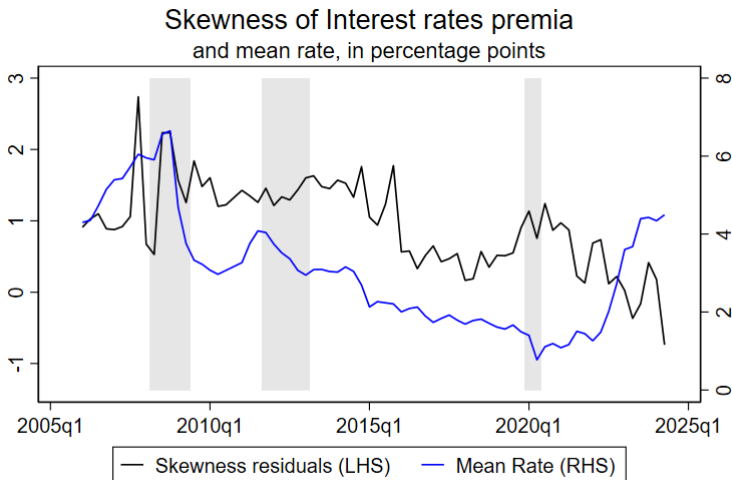
Source: Nouveaux Credits, BdF, 2006q1-2024q2
Excludes credit lines and personal loans.

Time variation



Source: Nouveaux Credits, BdF, 2006q1-2024q2
Excludes credit lines and personal loans.

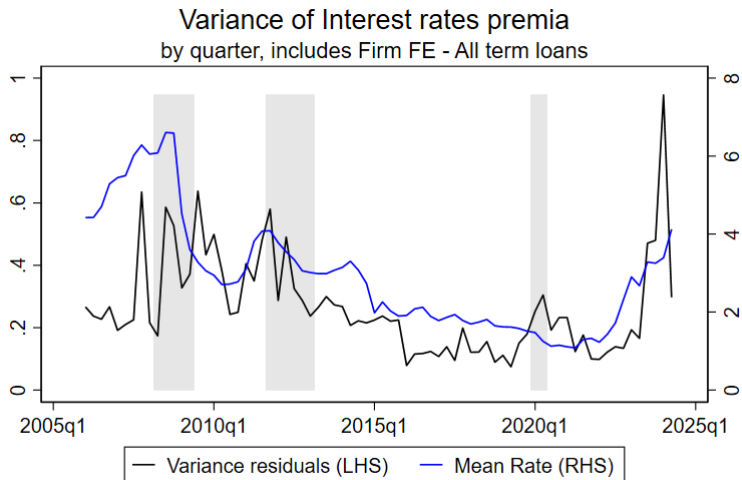
Skewness of rate premia



Source: Nouveaux Credits, BdF, 2006q1-2024q2. All term loans.

Recession dates for France from AFSE: 2008q2-2009q2, 2011q4-2013q1, 2020q1-2020q2

Variance of rate premia - Firm FE



Source: Nouveaux Credits, BdF, 2006q1-2024q2. Excludes credit lines and personal loans. Recession dates for France from AFSE: 2008q2-2009q2, 2011q4-2013q1, 2020q1-2020q2

Specification: firm fixed effects

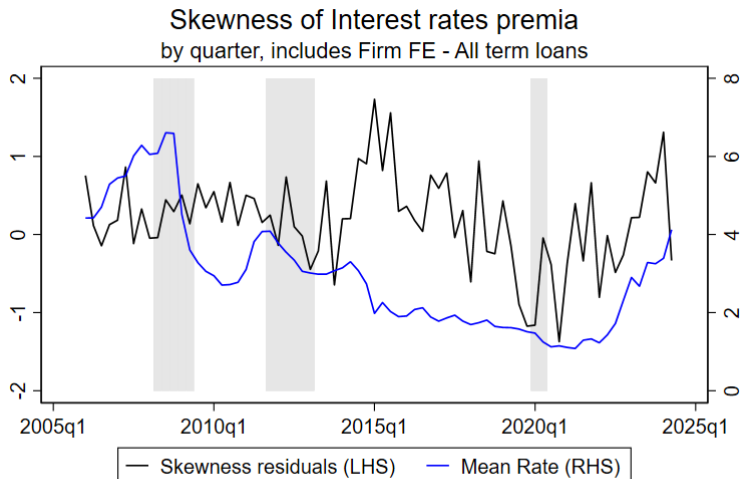
	Total effective rate					
Loan quantity (K-Eur)	-0.0000*** (0.000)	-0.0000*** (0.006)	-0.0000 (0.486)	-0.0000 (0.143)	-0.0000** (0.044)	-0.0000 (0.117)
Maturity (months)	-0.0016*** (0.000)	0.0013*** (0.000)	0.0015*** (0.000)	0.0015*** (0.000)	0.0014*** (0.000)	0.0014*** (0.000)
Variable rate (dummy)	0.2381*** (0.000)	0.1357*** (0.000)	0.2176*** (0.000)	0.2368*** (0.000)	0.2113*** (0.000)	0.1995*** (0.000)
Loan over assets	0.0524*** (0.000)	-0.0068** (0.025)	-0.0076*** (0.009)	-0.0070** (0.012)	-0.0069** (0.012)	-0.0074*** (0.007)
Quarter FE	✓					
Firm × Quarter FE		✓	✓	✓	✓	
Lender FE			✓			
Lender × Quarter FE				✓	✓	
Firm × Lender × Quarter FE						✓
Loan Type FE					✓	✓
Observations	590,621	197,343	197,328	196,048	196,042	156,211
R ²	0.51	0.92	0.93	0.94	0.94	0.94

p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

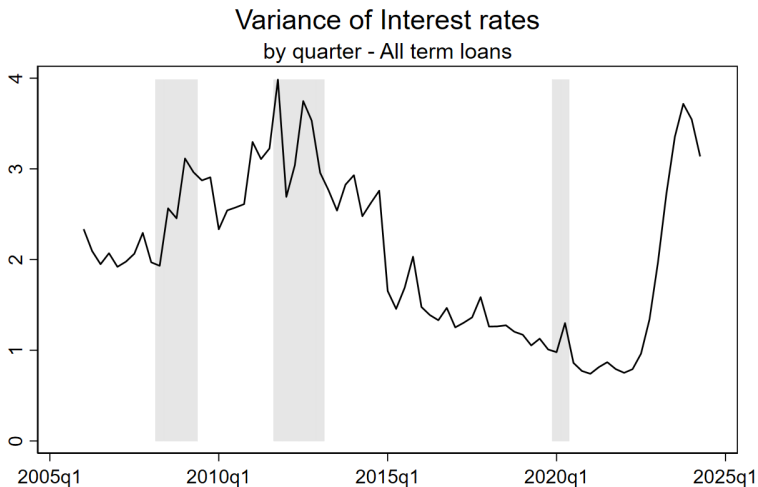
Table 6: "Data: Nouveaux Credits (BdF), 2006q1-2024q2"" Excludes credit lines and personal loans. All credit and"" balancesheet variables are expressed in terms of reported assets."

Skewness of rate premia



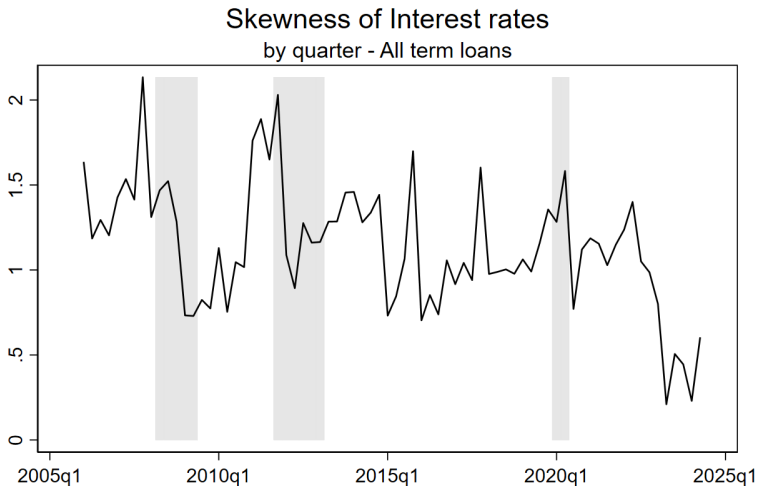
Source: Nouveaux Credits, BdF, 2006q1-2024q2. Excludes credit lines and personal loans. Recession dates for France from AFSE: 2008q2-2009q2, 2011q4-2013q1, 2020q1-2020q2

Distribution moments in the time series - Rates



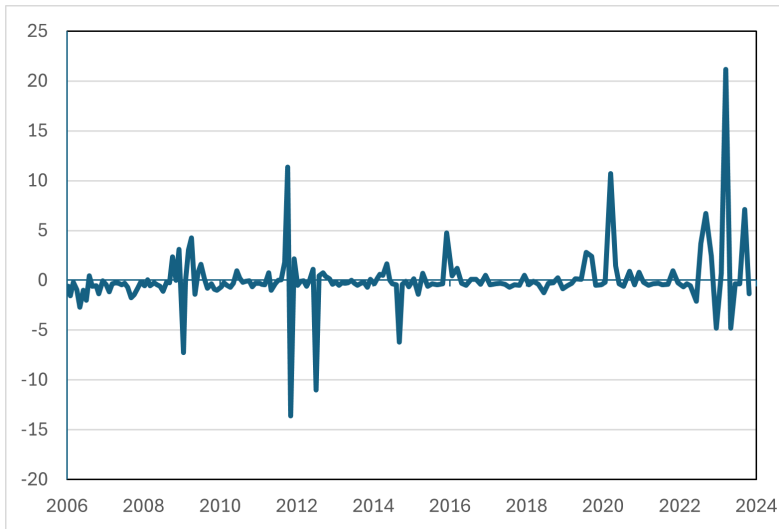
Source: Nouveaux Crédits, BdF, 2006q1-2024q2. Excludes credit lines and personal loans. Recession dates for France from AFSE: 2008q2-2009q2, 2011q4-2013q1, 2020q1-2020q2

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Monetary Policy shock



Effect of an increase of the policy rate

I compute and use high frequency monetary policy shocks to test the effect of an increase in the funding cost of banks

In levels:²

$$Y_{lender,t} = \beta_0 + \beta_1 \cdot \text{MPshock}_t + \beta_2 Y_{lender,t-1} + \text{FE}_{lender} + \varepsilon$$

And in first differences:³

$$\Delta Y_{lender,t} = \gamma_0 + \gamma_1 \cdot \text{MPshock}_t + \text{FE}_{lender} + \epsilon$$

►► shock series

² β_1 measures the effect of MP on the level of dispersion of premia, controlling for its lag. Assumes that there is inertia in the dispersion of rates over time.

³ γ_1 measures the effect of MP on the change of dispersion of premia. No need to worry about trend.

Effect of an increase of the policy rate

	sandard deviation		delta SD	
L1.rate shock	-0.0096*** (0.000)	-0.0085*** (0.003)	-0.0071** (0.028)	-0.0049 (0.175)
L2.rate shock		0.0001 (0.979)		0.0069** (0.041)
L3.rate shock		-0.0017 (0.568)		-0.0007 (0.851)
L1.standard deviation	0.2813*** (0.000)	0.2799*** (0.000)		
Lender FE	✓	✓	✓	✓
Observations	7,189	7,077	7,189	7,077
R^2	0.40	0.41	0.01	0.01

p-values in parentheses, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Data: Nouveaux Credits, BdF, 2006q1-2024q2, quarterly shocks. The table shows the effect of a monetary policy rate shock on the standard deviation of the distribution.

Effect of an increase of the policy rate

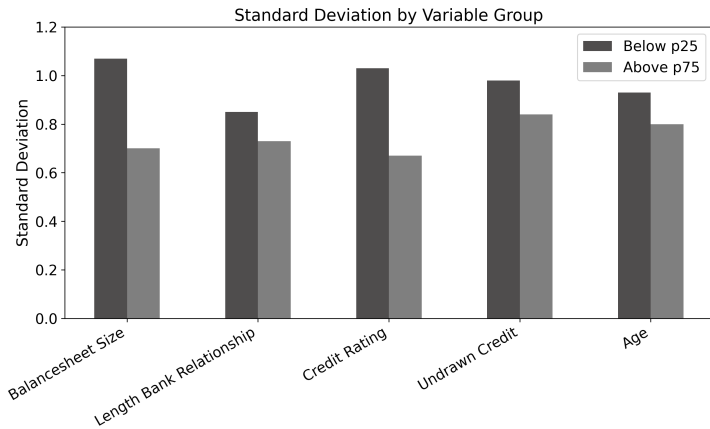
	skewness		delta skewness	
L1.rate shock	-0.0135*** (0.010)	-0.0099* (0.093)	0.0012 (0.860)	0.0053 (0.490)
L2.rate shock		-0.0227*** (0.000)		-0.0024 (0.732)
L3.rate shock		-0.0208*** (0.000)		-0.0099 (0.196)
L1.skewness	0.1294*** (0.000)	0.1171*** (0.000)		
Lender FE	✓	✓	✓	✓
Observations	7,189	7,077	7,189	7,077
R ²	0.23	0.24	0.00	0.00

p-values in parentheses, * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Data: Nouveaux Credits, BdF, 2006q1-2024q2, quarterly shocks. The table shows the effect of a monetary policy rate shock on the skewness of the distribution

Dispersion: heterogeneity over firm characteristics

Information asymmetries and imperfect competition are weaker for better firms

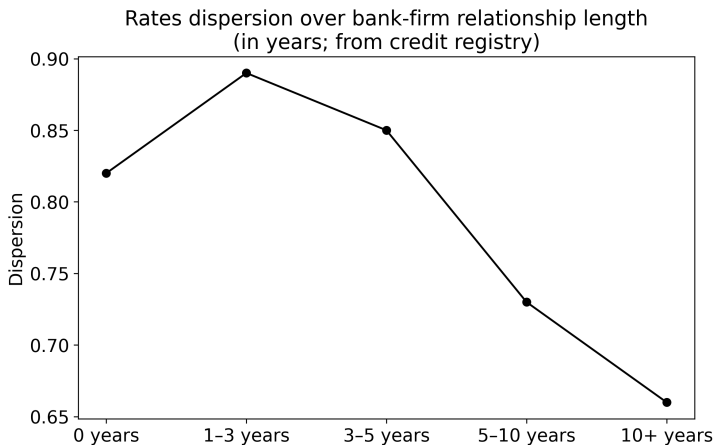


Dispersion: descriptives

Rate dispersion - descriptives							
Variables		below p25	above p75	Variables		below p25	above p75
balancesheet	s.d. n	1.07 107	0.70 119	undrawn credit	s.d. n	0.98 138	0.84 113
turnover over assets	s.d. n	0.74 107	0.93 123	relationship length	s.d. n	0.85 41	0.73 81
credit rating	s.d. n	0.67 74	1.03 73	distance last review	s.d. n	0.90 107	0.82 4
age	s.d. n	0.93 25	0.80 192	pledged guarantees	s.d. n	0.83 264	0.93 193
maturity	s.d. n	1.03 202	0.69 26	short term debt	s.d. n	0.71 106	0.92 124
loan over assets	s.d. n	0.83 118	0.93 114		s.d. n		

Table 9: Number of observations in thousands.

Dispersion over the bank-firm relationship

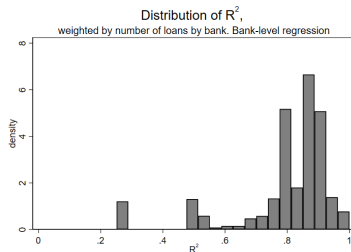
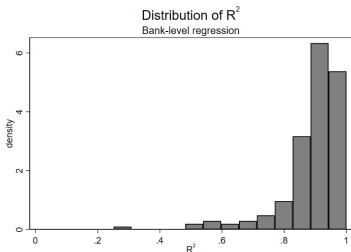


Source: Nouveaux Crédits and Credit Registry.

Within or Between bank dispersion?

Is the dispersion equal across banks, or are some banks "contributing more" to observed dispersion?

- I re-run the analysis separately for all banks and collect R^2



The right image reports the distribution of R^2 weighted by the number of loans issued.

Default frequencies by rating

3-year ahead default frequencies, by rating class and year (2004-2022).

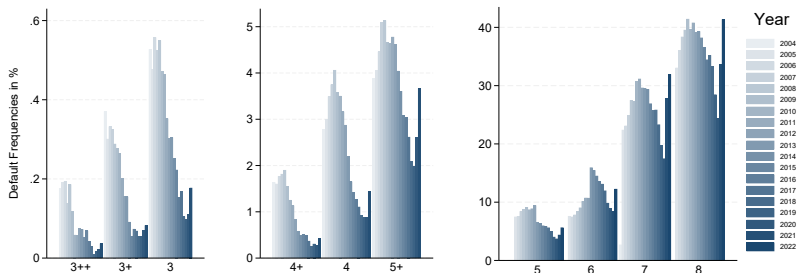


Figure 5: Source: COTA (ratings by BdF), 2004-2025. Following the same approach by BdF in published reports, I consider all firms with a rating at 01/01/year and compute the default frequencies at 31/12/year+2, by year and rating class.

BdF Credit Rating

Three possible entries.

1. Credit Rating (= Probability of default)

Rating Category	3++	3+	3	4+	4	5+	5	6	7	8	9
Default Rate	0.02%	0.05%	0.13%	0.35%	0.96%	2.38%	6.30%	9.24%	21.58%	36.98%	36.98%

Table 10: Correspondence of rating categories and default probabilities
(Source: BdF)

2. Actual Default

Takes value "P"

3. Not enough information

Takes value "0"

Banque de France rating scale

LA COTE DE CRÉDIT

(capacité de
l'entreprise
à honorer ses
engagements
financiers à 3 ans)

3++	Excellente
3+	Très forte
3	Forte
4+	Assez forte
4	Correcte
5+	Assez faible
5	Faible
6	Très faible
7	Appelant une attention spécifique présence d'au moins un incident de paiement significatif
8	Menacée
9	Compromise
P	Procédure collective redressement ou liquidation judiciaire
0	Pas de documentation comptable analysée et absence d'informations défavorables

Forward-looking over three years; combines accounting data, qualitative information and payment incidents.

Can rating be predicted?

	Rating change - 3y ahead				
Interest rate	0.0091***	0.0044	-0.0008	-0.0014	0.0235***
Loan Amount	0.0049*	0.0019	0.0015	0.0052	0.0045
Maturity (months)	0.0000	0.0000	0.0001	0.0002**	0.0002**
Age	-0.0003	-0.0004	-0.0005	-0.0004	-0.0010***
Revenues	-0.0068***	-0.0130***	-0.0131***	-0.0135***	-0.0190***
Bank Debt	-0.1588***	-0.1584***	-0.1552***	-0.1538***	0.0005
Non-Bank Debt	-0.2235***	-0.1952***	-0.1984***	-0.2021***	0.0434*
Short Term Debt	0.0108	0.0301	0.0323	0.0308	0.3190***
Marge disponible crédits confirmés (KEUR)	-0.0147***	-0.0176***	-0.0186***	-0.0181***	-0.0218***
Bank risk weighting	0.0039***	0.0027**	0.0023	0.0025	0.0023
Quarter FE	✓				
Lender × Quarter FE		✓			
Lender × Quarter × 2digit Sector FE			✓		
Lender × Quarter × Loan Type × 2digit Sector FE				✓	✓
Region FE					✓
Group indicator, Rating FE					✓
Observations	306,794	306,794	306,794	306,794	306,794
R ²	0.01	0.06	0.22	0.27	0.34

p-values in parentheses

* p < 0.10, ** p < 0.05, *** p < 0.01

Table 11: Data: Nouveaux Credits, Credit Registry (BdF), 2006q1-2024q2. Excludes credit lines and personal loans. All credit and balancesheet variables are expressed in terms of reported assets.

Which banks use the BdF credit rating?

I compute the δR^2 explained by the credit rating.

- I re-run the analysis separately for all banks, controlling only for time and loan type FEs, and collect the δR^2

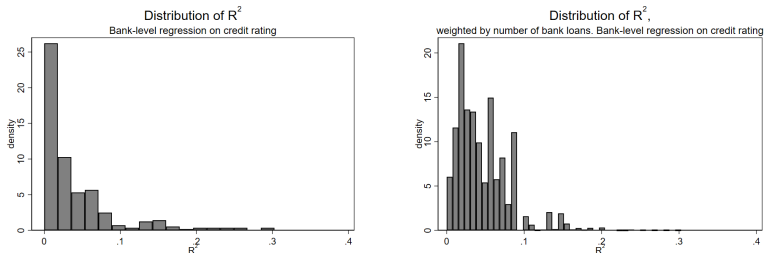


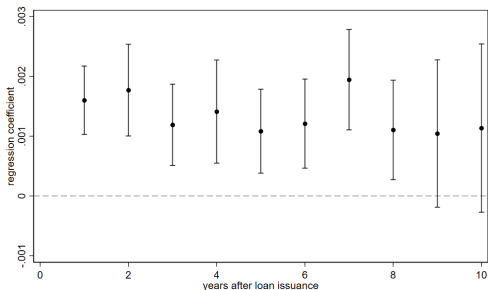
Figure 6: The right image reports the distribution of R^2 weighted by the number of loans issued.

Correlation residuals vs actual default

Same test, replacing the rating revision with the default itself:

$$r_{ilbt} = \alpha + \sum_{j=1}^k \beta_j^D D_{i,t+j} + \mathbf{x}'_{ilbt} \gamma + \mu_{bt\tau s} + \varepsilon_{ilbt}$$

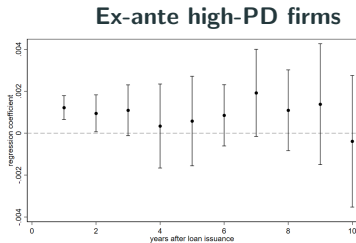
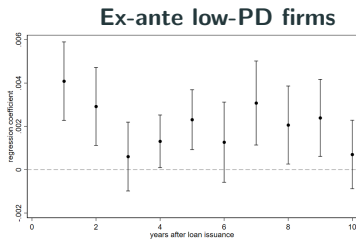
$D_{i,t+j} = 100$ if firm i defaults in year j after issuance. Estimated once per horizon $k = 1, \dots, 10$; needs no balanced panel; clustered by lender and by quarter.



►► by rating

►► 1y prob default

Actual default: by ex-ante rating



Computed separately for firms with ex-ante $PD < 0.62\%$ vs $PD > 3.45\%$.

Robustness: maturity $\geq$ 24 months

	Default Probability - 2021q3-2021q4 only						
Variable rate (dummy)	-0.5698 (0.169)		-19.6243 (0.479)	-0.5121 (0.186)			
annual dRate end y2			5.0816 (0.477)		0.0935 (0.943)	0.6439 (0.316)	
annual dRate end y3			2.6790 (0.563)		-0.7103 (0.718)	-1.5704 (0.179)	
Variable rate (dummy) $\times$ Loan over assets				-0.6271 (0.249)			
annual dRate end y2 $\times$ Loan over assets					5.0753 (0.781)		
annual dRate end y2 $\times$ Bank Debt share							-0.6506 (0.424)
annual dRate end y3 $\times$ Loan over assets					-7.6615 (0.769)		
annual dRate end y3 $\times$ Bank Debt share							0.9643 (0.155)
Loan over assets	-0.5329 (0.374)	-1.1153 (0.168)	-1.1298 (0.233)	-0.1584 (0.740)	-0.6351 (0.216)	-0.6550 (0.287)	-1.1379 (0.283)
Bank Debt share	0.0986 (0.725)	-0.2617 (0.313)	-0.2685 (0.569)	0.1401 (0.631)	-0.2242 (0.369)	-0.2431 (0.431)	-0.2795 (0.674)
Lender $\times$ Month $\times$ Loan Type $\times$ 2digit Sector FE	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓
All other observables	✓	✓	✓	✓	✓	✓	✓
Observations	4,324	4,089	4,089	4,324	4,089	4,089	4,089
R ²	0.41	0.41	0.41	0.41	0.41	0.41	0.41

p-values in parentheses

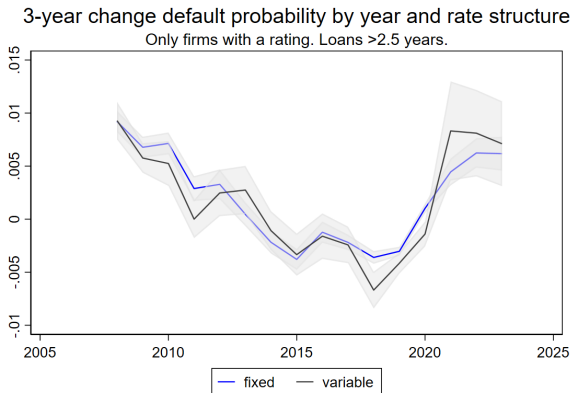
* p < 0.10, ** p < 0.05, *** p < 0.01

Table 12: Data: Nouveaux Crédits (BdF), 2006q1-2024q2. Term loans, maturity at least 24 months, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

Validity checks

- **Mitigate self-selection issue:**
 - contract choice is predominantly driven by supply-side factors
 - end of ZLB period, firms do not expect a rate hike
- **Balance table (in 2021):**
 - some characteristics differ (variable: fewer firms, larger size, older, riskier, larger financial debt)
 - All of those are included as controls
- **Parallel trends test:**
 - visually: rating transitions for firms taking $> 2.5y$ loans do not differ significantly [▶▶ Figure](#)
 - Pre-treatment betas are close to zero (not ***)

Parallel trends: fixed vs. floating



Mean three-year change in the default probability by year, for fixed- and floating-rate borrowers separately. Term loans over 2.5 years, firms with a rating. Shaded: 95% confidence bands.

The two groups track each other through the pre-period.

Can variable/fixed rate contract be predicted?

Focusing on firms with a rating at the time of the loan, I estimate:

$$Floating_{loan} = \beta X^{observables} + \delta_{t,b,p,s} + \varepsilon_{loan} \quad (2)$$

where $Floating_{loan}$ is a dummy for variable-rate loans and $\delta_{t,b,p,s}$ is the fixed effects interaction term for month, bank, loan purpose and firm sector.

Table 13: Variance decomposition of variable rate choice

Fixed Effects Included	Adjusted R^2
Quarter	0.04
+ Lender	0.63
+ Loan Type	0.86
+ Firm Sector	0.91

the contract choice is predominantly driven by supply-side and sectoral factors rather than idiosyncratic firm risk

Dispersion over PD

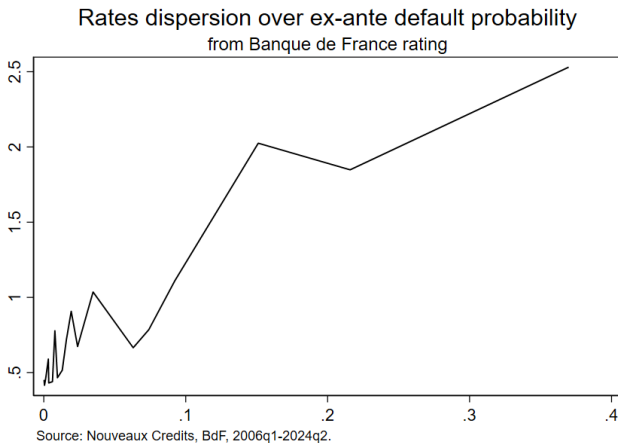


Figure 7: Standard deviation of errors from baseline regression, by probability of default buckets

Theory

Dispersed rate equilibrium: Intuition

Banks choose the equilibrium **rate distribution** that maximizes expected profits.

Limit cases of $\pi = 0, 1$

- Banks operate under monopoly/perfect competition
→degenerate distribution

Intermediate case $\pi \in (0, 1)$

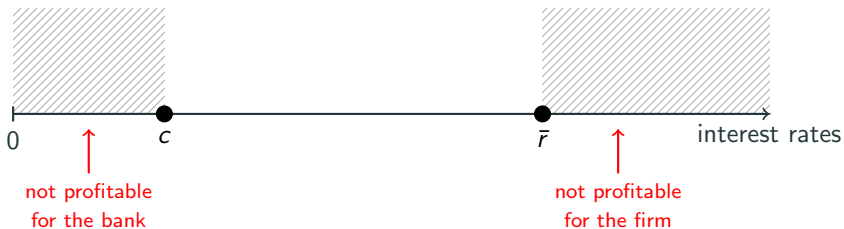
- Trade-off between higher profits/lower acceptance probability
→non-degenerate distribution

$$Profits^i(r) = \begin{cases} \mu_i[(1 - \pi) + \pi(1 - F(r))](r - c_i) & \text{if } r \leq \bar{r} \\ 0 & \text{otherwise} \end{cases}$$

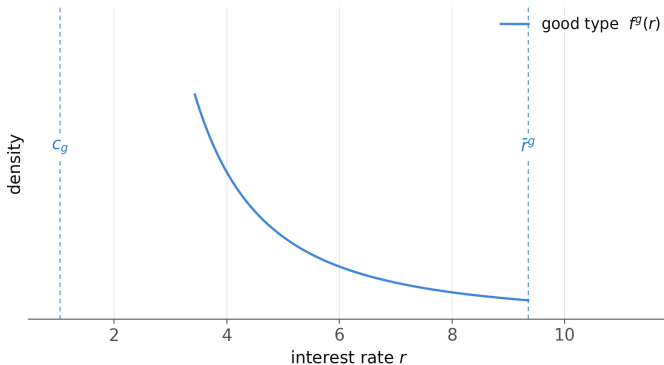
→ Find the **isoprofit function** that maximizes expected profits.

Search frictions only

Example: feasible rates



Example: search frictions generate within-type dispersion



- Bank profits: $\Pi(r) = (1 - \pi + \pi(1 - F(r)))(r - c)$
- $\pi \rightarrow 0$ (monopoly) $r^* = \bar{r}$, $\pi \rightarrow 1$ $r^* = c$ (Bertrand)
- for $\pi \in (0, 1)$: mixed strategy equilibrium

Equilibrium under a perfect signal

Proposition 4 (Dispersed-rate equilibrium, $q = 1$)

For $\pi \in (0, 1)$ and a perfect signal there is a unique symmetric mixed-strategy equilibrium. For each type i , banks randomize over $[\underline{r}^i, \bar{r}^i]$ according to

$$F^i(r) = 1 - \frac{1 - \pi}{\pi} \left[\frac{\bar{r}^i - r}{r - c_i(r)} \right],$$

with $\bar{r}^i$ the type's reservation rate and $\underline{r}^i$ defined by $F^i(\underline{r}^i) = 0$.

Observed rates are the mixture $\mu_g F^g(r) + \mu_b F^b(r)$ on $[\underline{r}^g, \bar{r}^b]$.

- the bank solves **two separate problems**, one per type $\rightarrow$ *between-type* dispersion

►► Derivation

►► Example: one type

►► Example: two types

Perfect signal: why banks mix

With $q = 1$ the bank solves one problem per type. Its expected profit on a quote r to type i :

$$Profits^i(r) = \begin{cases} \mu_i[(1 - \pi) + \pi(1 - F^i(r))](r - c_i) & \text{if } r \leq \bar{r}^i \\ 0 & \text{otherwise} \end{cases}$$

Trade-off: probability of acceptance and profits per trade

No pure strategy survives:

- at a common rate $r > c_i$ a bank gains by undercutting — rates fall to c_i
- at c_i a bank gains by quoting $\bar{r}^i$ to its $1 - \pi$ captive firms

→ **mixed strategy equilibrium** over $[\underline{r}^i, \bar{r}^i]$

Dispersed rate equilibrium $F(r)$

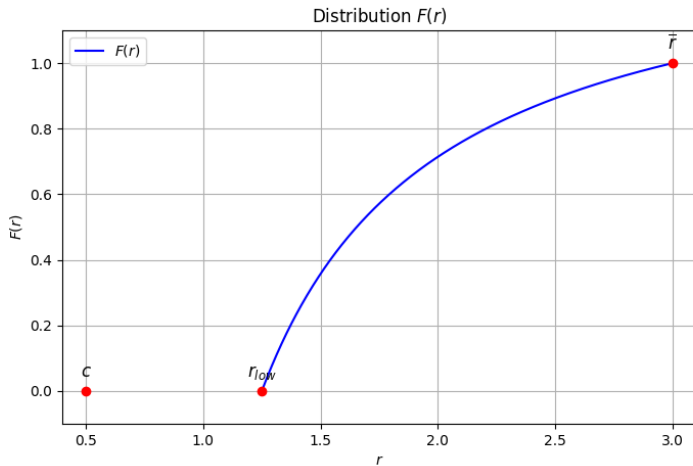


Figure 8: Rates distribution - one type

ϕ : how severe is adverse selection?

Lemma 2 (Monopoly benchmark)

Under $\pi = 0$, $c'(r) < 1$ and an uninformative signal, define

$$\phi \equiv (\bar{r}^g - c^{avg}(\bar{r}^g)) - \mu_b(\bar{r}^b - c_b(\bar{r}^b))$$

The unique equilibrium is $\bar{r}^g$ if $\phi > 0$, and $\bar{r}^b$ if $\phi \leq 0$.

- $\phi > 0$: adverse selection is **mild** — the bank earns more by **pooling** both types
- $\phi \leq 0$: adverse selection is **severe** — the bank earns more from **bad types only**, and good types walk away

Lemma 3 (Perfect competition)

Under $\pi = 1$ the unique equilibrium is a pooling contract at the competitive rate $r^c = c^{avg}$, accepted by the whole population; profits are zero.

No Info - Monopoly

If monopoly $\pi = 0$

$$Profits = \begin{cases} (r - c^{avg}) & \text{if } \underline{r} < r \leq \bar{r}^g \\ \mu_b(r - c_b) & \text{if } \bar{r}^g < r \leq \bar{r}^b \end{cases}$$

The bank evaluates what the maximum profit it can extract in each region is. It compares:

$$\max\{(1 - \pi)(\bar{r}^g - c_{avg}), \mu_b(1 - \pi)(\bar{r}^b - c_b)\}$$

Define ϕ :

$$\phi = (\bar{r}^g - c_{avg}) - \mu_b(\bar{r}^b - c_b)$$

1. $\phi \leq 0$ (strong AS) $\rightarrow$ Banks extract more profits from bad types only
2. $\phi > 0$ (weak AS) $\rightarrow$ Banks extract more profits from pooling good and bad types

No Info - Perfect competition

If perfect competition $\pi = 1$

1. Banks make zero profits, offering the average cost in the population
 $c^{avg} = \mu_g c_g + \mu_b c_b$
2. As I consider the case $c^{avg} < \bar{r}_g$, the pooling rate will be always accepted by both types

$$Profits = (\mu_b + \mu_g)(c^{avg} - c^{avg}) = 0$$

If p is a decreasing function of r

$$p = p(r) = \bar{p}^i - \gamma r$$

Case 1: γ is sufficiently small

- Firm: cares about profit if success $R-r$, same $\bar{r}$
- Bank requires compensation for risk $\uparrow c = \frac{r^{CB}}{p(r)}$

The equilibrium distribution $F(r)$:

$$F(r) = 1 - \frac{1 - \pi}{\pi} \frac{\bar{r} - r}{r - \frac{r^{CB}}{p(r)}}, \quad \text{if } \underline{r(\pi)} < r \leq \bar{r}$$

$$\underline{r(\pi)} = (1 - \pi) \left(\bar{r} - \frac{r^{CB}}{p(r)} \right) + \frac{r^{CB}}{p(r)}$$

If p is a decreasing function of r

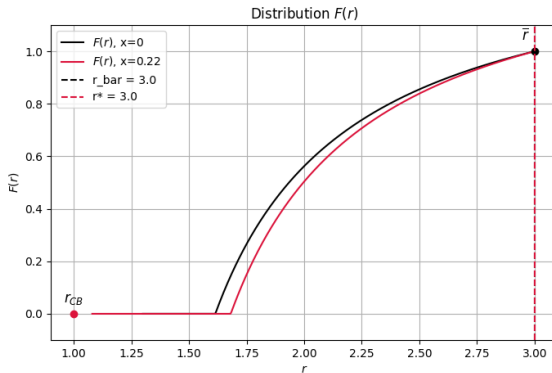


Figure 9: Example: $\bar{r} = 3$, $\pi = 0.7$, $r^{cb} = 1$, $p^{success} = 0.98 - x * r$

, values of $x = [0, 0.05]$.

If p is a decreasing function of r

Case 2: γ is sufficiently high

- As before, banks require compensation for risk $\rightarrow \uparrow c$
- $p(r)$ may go to zero if r is too high $\rightarrow \bar{r} < R$
- convex cost of lending $\rightarrow$ Interior rate r^* that maximizes profits

$\rightarrow$ The interval of feasible rates is smaller

Equilibrium distribution $F(r)$:

$$F(r) = 1 - \frac{1 - \pi}{\pi} \frac{r^* - r}{r - \frac{r^{CB}}{p(r)}}, \quad \text{for } \underline{r}(\pi) < r \leq \bar{r}$$

If p is a decreasing function of r

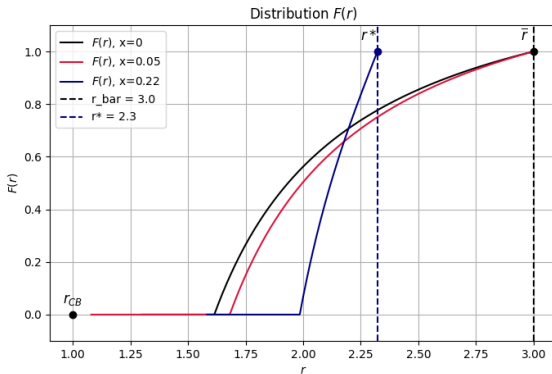


Figure 10: Example: $\bar{r} = 3$, $\pi = 0.7$, $r^{cb} = 1$, $p^{success} = 0.98 - \gamma * r$, values of $\gamma = [0, 0.05, 0.22]$.

Note that the correlation between rates and ex-post default probability is γ .

Perfect vs asymmetric information

1. **Perfect information / perfect signal $q=1$:**

- The bank can perfectly identify each firm's type
- Positive correlation between rates and riskiness

2. **Asymmetric information / $q=0.5$**

- The bank cannot distinguish types

3. The bank can **acquire costly information** $q \in 0.5, 1$ about firm types

- extreme precision q_θ values coincide with no/perfect information cases

Perfect information - Total CDF

Duplicate the one-firm problem. $F(r)$ is now the sum of both distribution

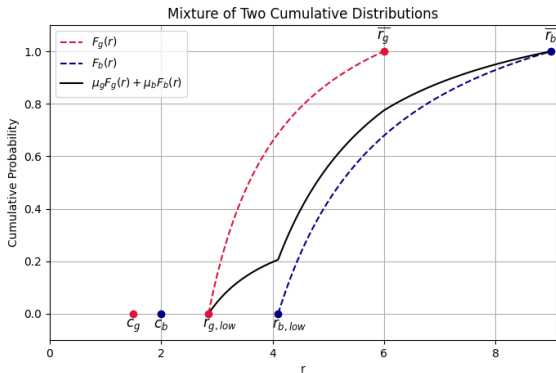


Figure 11: Simulation. Parameter values $c_g = 1.5$, $c_b = 2$, $\bar{r}^g = 6$, $\bar{r}^b = 9$, $\pi = 0.7$, $\mu_b = 0.7$

Perfect information - Density

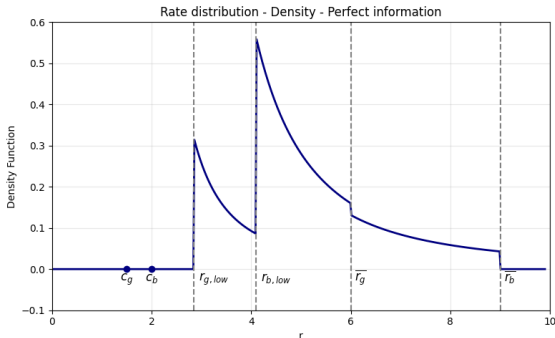
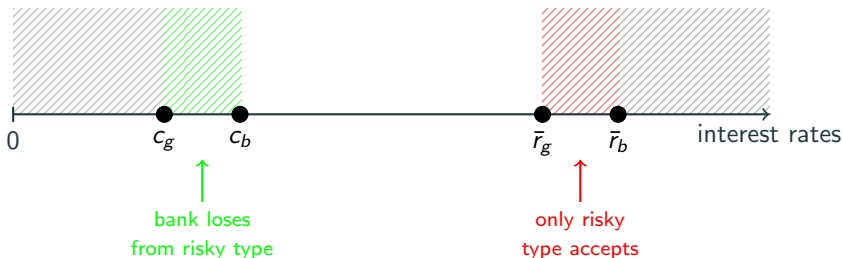


Figure 12: Simulation. Parameter values $c_g = 1.5$, $c_b = 2$, $\bar{r}^g = 6$, $\bar{r}^b = 9$, $\pi = 0.7$, $\mu_b = 0.7$

$$f^i(r) = \begin{cases} \frac{1 - \pi}{\pi} \frac{\bar{r}^i - c_i}{(r - c_i)^2}, & \text{if } \underline{r}^i(\pi) < r \leq \bar{r}^i, \\ 0, & \text{otherwise.} \end{cases}$$

Asymmetric information

Example: feasible rates



- Hp: risky type has higher default probability but higher reservation rate (delivers consistent correlation patterns)
- the bank cannot perfectly recognize types, but knows μ_{bad} , π , $\bar{r}_{bad}$, $\bar{r}_{good}$, p_g , p_b

Asymmetric information

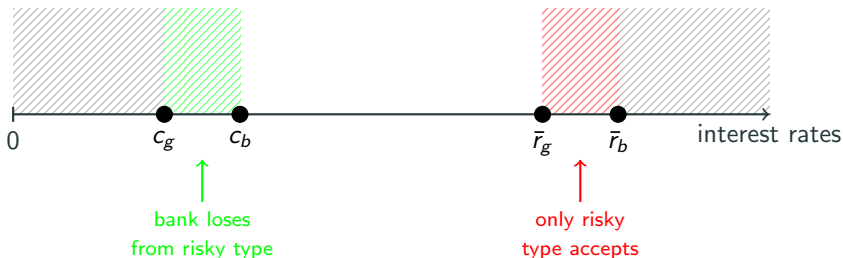
Example: feasible rates



- Hp: bad type has higher default probability but higher reservation rate (delivers consistent correlation patterns)
- the bank cannot perfectly recognize types, but knows μ_{bad} , π , $\bar{r}_{bad}$, $\bar{r}_{good}$, p_g , p_b

Asymmetric information

Example: feasible rates



- Hp: risky type has higher default probability but higher reservation rate (delivers consistent correlation patterns)
- the bank cannot perfectly recognize types, but knows μ_{bad} , π , $\bar{r}_{bad}$, $\bar{r}_{good}$, p_g , p_b

Equilibrium without information: two regimes

Proposition 5 (Severe adverse selection, $\phi \leq 0$)

Banks randomize over $[\underline{r}^a, \bar{r}^b]$, mixing a pooling and a bad-only region:

$F = F^a$ on $[\underline{r}^a, \bar{r}^g]$ and $F = F^a(\bar{r}^g) + F^b$ on $(\bar{r}^g, \bar{r}^b]$, with

$$F^a(r) = \max \left\{ 1 - \frac{1 - \pi}{\pi} \left[\frac{\mu_b(\bar{r}^b - c_b)}{r - c^{avg}(r)} - 1 \right], 0 \right\}$$

$$F^b(r) = 1 - F^a(\bar{r}^g) - \frac{1 - \pi}{\pi} \left[\frac{\bar{r}^b - c_b}{r - c_b(r)} - 1 \right]$$

Proposition 6 (Mild adverse selection, $\phi > 0$)

Banks randomize over $[\underline{r}, \bar{r}^g]$ only, at

$F(r) = 1 - \frac{1 - \pi}{\pi} [(\bar{r}^g - r)/(r - c^{avg}(r))]$: all firms trade, both types face the same rates.

Lemma 4 (When does pooling survive?)

A pooling region exists for $-\frac{\pi}{1 - \pi}(\bar{r}^g - c^{avg}) < \phi \leq 0$; below that threshold the bank lends to bad types only.

Case 1. Banks extract more profits at $\bar{r}^b$.

The **CDF** on both regions can be expressed as:

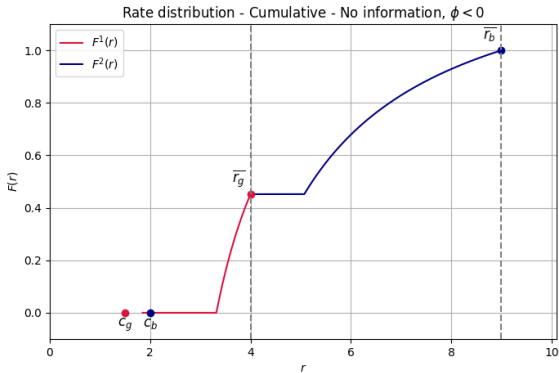
$$F(r) = \begin{cases} 0 & \text{if } r < \underline{r}^a \\ F^a(r) & \text{if } \underline{r}^a \leq r \leq \bar{r}^g \\ F^a(\bar{r}^g) + F^b(r) & \text{if } \bar{r}^g < r \leq \bar{r}^b \\ 1 & \text{if } r > \bar{r}^b \end{cases}$$

Then, the **equal profit condition**:

$$[1 - \pi + \pi(1 - F^a(r))](r - c^{avg}) = \mu_b(1 - \pi)(\bar{r}^b - c_b)$$

$$\mu_b[1 - \pi + \pi(1 - F^a(\bar{r}^g) - F^b(r))](r - c_b) = \mu_b(1 - \pi)(\bar{r}^b - c_b)$$

No info - $\phi \leq 0$

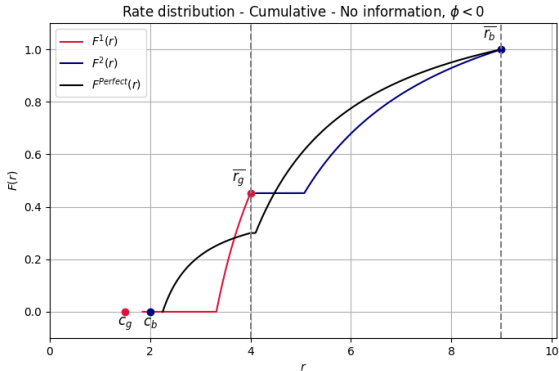


Parameter values: $c_g = 1.5$, $c_b = 2$, $\bar{r}_g = 6$, $\bar{r}_b = 9$, $\pi = 0.7$, $\mu_b = 0.7$

$$F^a(r) = 1 - \frac{1 - \pi}{\pi} \left[\frac{\mu_b(\bar{r}^b - c_b)}{(r - c^{avg})} - 1 \right]$$

$$F^b(r) = 1 - F^a(\bar{r}_g) - \frac{1 - \pi}{\pi} \left[\frac{(\bar{r}^b - c_b)}{(r - c_b)} - 1 \right]$$

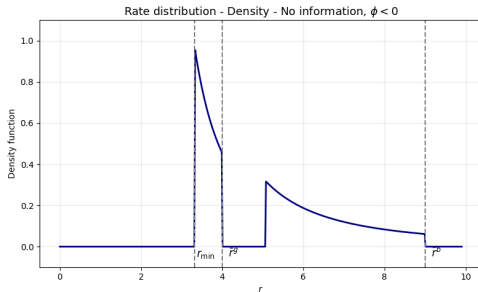
No info - $\phi \leq 0$



Parameter values $c_g = 1.5$, $c_b = 2$, $\bar{r}_g = 4$, $\bar{r}_b = 9$, $\pi = 0.7$, $\mu_b = 0.7$

More information, higher dispersion

No info - $\phi \leq 0$



Parameter values $c_g = 1.5$, $c_b = 2$, $\bar{r}^g = 4$, $\bar{r}^b = 9$, $\pi = 0.7$, $\mu_b = 0.7$

$$f(r) = \begin{cases} \frac{1 - \pi}{\pi} \frac{\mu_b(\bar{r}^b - c_b)}{(r - c^{avg})^2}, & \text{if } \underline{r} \leq r \leq \bar{r}^g, \\ \frac{1 - \pi}{\pi} \frac{(\bar{r}^b - c_b)}{(r - c_b)^2}, & \text{if } \underline{r}^b < r \leq \bar{r}^b, \\ 0, & \text{otherwise.} \end{cases}$$

Case 2. Banks extract more profits at $\bar{r}^g$.

Intuition: No contract on bad-types-only region is feasible. Full pooling equilibrium.

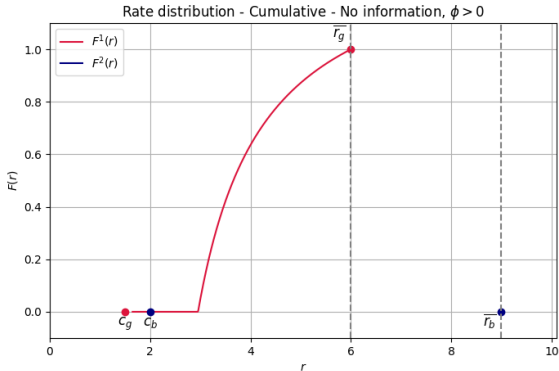
The **CDF** on both regions can be expressed as:

$$F(r) = \begin{cases} 0 & \text{if } r < \underline{r}^a \\ F^a(r) & \text{if } \underline{r}^a \leq r \leq \bar{r}^g \\ 1 & \text{if } r > \bar{r}^g \end{cases}$$

Then, the **equal profit condition**:

$$[1 - \pi + \pi(1 - F^a(r))](r - c^{avg}) = (1 - \pi)(\bar{r}^g - c^{avg})$$

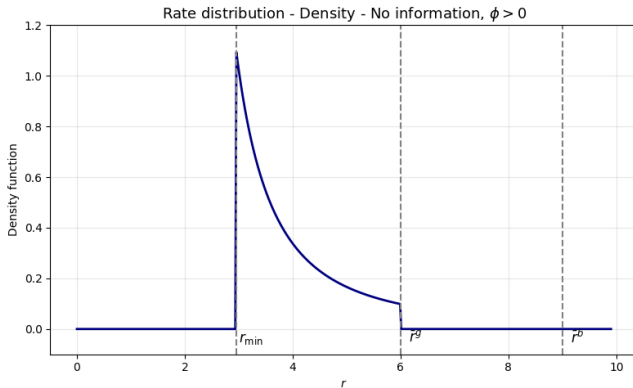
No info - $\phi > 0$



Parameter values $c_g = 1.5$, $c_b = 2$, $\bar{r}^g = 6$, $\bar{r}^b = 9$, $\pi = 0.7$, $\mu_b = 0.3$

$$F^a(r) = 1 - \frac{1 - \pi}{\pi} \left[\frac{(\bar{r}^g - r)}{(r - c^{avg})} \right]$$

No info - $\phi > 0$

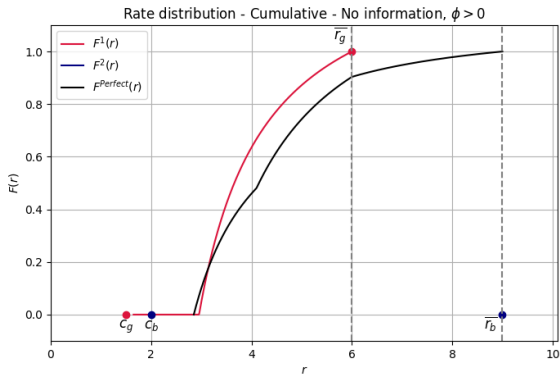


Parameter values $c_g = 1.5$, $c_b = 2$, $\bar{r}^g = 6$, $\bar{r}^b = 9$, $\pi = 0.7$, $\mu_b = 0.3$

$$F^a(r) = 1 - \frac{1 - \pi}{\pi} \left[\frac{(\bar{r}^g - r)}{(r - c^{avg})} \right]$$

No info - $\phi > 0$

Figure 13: Compare pooling eq. (red) and Perfect info eq. (black)



Parameter values $c_g = 1.5, c_b = 2, \bar{r}_g = 6, \bar{r}_b = 9, \pi = 0.7, \mu_b = 0.3$

More information, higher dispersion

Acquire Information

The bank observes the signal s and updates the belief (unconditional probability) using Bayes' rule.

Conditional probabilities: By Bayes' rule, the probability that a firm is good given a good signal is

$$\begin{aligned}\Pr(\theta = G \mid s = G) &= \frac{\Pr(\theta = G \cap s = G)}{\Pr(s = G)} \\ &= \frac{\Pr(s = G \mid \theta = G) \Pr(\theta = G)}{\Pr(s = G \mid \theta = G) \Pr(\theta = G) + \Pr(s = G \mid \theta = B) \Pr(\theta = B)} \\ &= \frac{q_g \mu_g}{q_g \mu_g + (1 - q_b) \mu_b}\end{aligned}$$

Signal Precision & Bayesian Updating

Signal precision: For each type $\theta \in \{g, b\}$, the bank observes a binary signal s with

$$\Pr(s = \theta \mid \theta) = q_\theta \geq 0.5$$

$q_\theta = \frac{1}{2}$: uninformative (no information); $q_\theta = 1$: perfect information.

Unconditional signal probabilities:

$$\sigma_g \equiv \Pr(s = g) = q_g \mu_g + (1 - q_b) \mu_b, \quad \sigma_b \equiv \Pr(s = b) = q_b \mu_b + (1 - q_g) \mu_g$$

Bayesian updating:

$$\alpha^{s_g} \equiv \Pr(\theta = g \mid s = g) = \frac{q_g \mu_g}{q_g \mu_g + (1 - q_b) \mu_b},$$

$$\alpha^{s_b} \equiv \Pr(\theta = g \mid s = b) = \frac{(1 - q_g) \mu_g}{(1 - q_g) \mu_g + q_b \mu_b}$$

$$\alpha^{s_g} > \mu_g > \alpha^{s_b}$$

Acquire Information

Expected outcome:

1. Following the same procedure as before, banks compare profits at the two reservation prices using the updated probabilities
2. now, conditional on a good signal, the average population of good types met by the bank is higher than μ_g , and vice-versa for bad signals
3. in eq, two distribution, one for each signal realization

Outcome:

- $\uparrow$ informative signal $\rightarrow \downarrow$ adverse selection problem, because the average cost and population share are closer to real one
- $\uparrow$ informative signal $\rightarrow \uparrow$ dispersion
- $\uparrow$ informative signal $\rightarrow \uparrow$ correlation between en-ante rate and ex-post default probability

One type: equilibrium distribution

Equal-profit condition (highest rate must be the reservation rate $\bar{r}$):

$$(1 - \pi)(\bar{r} - c) = [1 - \pi + \pi(1 - F(r))](r - c)$$

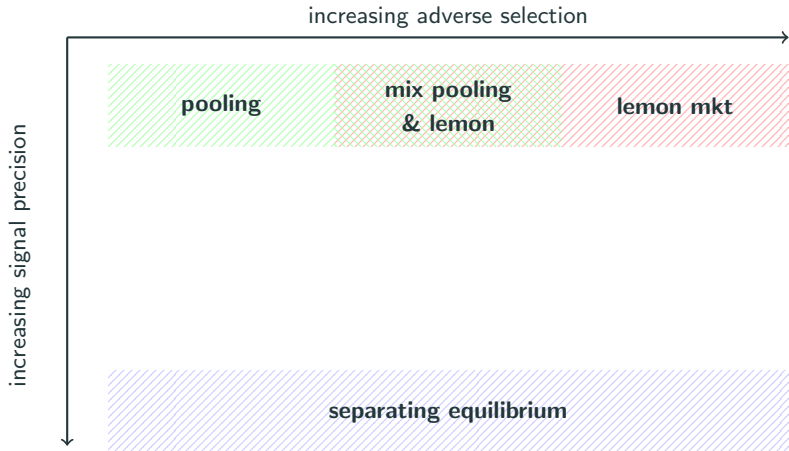
CDF and density:

$$F(r) = 1 - \frac{1 - \pi}{\pi} \left[\frac{\bar{r} - r}{r - c} \right], \quad f(r) = \frac{1 - \pi}{\pi} \frac{\bar{r} - c}{(r - c)^2}$$

on the support $\underline{r} \leq r \leq \bar{r}$, with $\underline{r} = (1 - \pi)\bar{r} + \pi c$ from $F(\underline{r}) = 0$.

$f'(r) < 0$: the density is right-skewed — most quotes cluster near the competitive rate, but captive firms sustain a tail up to $\bar{r}$.

Equilibrium: Overview



Gains from moving to a perfect signal

Bank, $\Delta\Pi \equiv \Pi^{q=1} - \Pi^{q=0.5}$, per firm approached by one bank:

Regime at $q = 0.5$	$\Delta\Pi$	the gain is the surplus ...
Full pooling, $\phi > 0$	$(1 - \pi) \mu_b (\bar{r}^b - \bar{r}^g)$	... not extracted from <i>bad</i> types
Hybrid, $\underline{\phi} < \phi \leq 0$	$(1 - \pi) \mu_g (\bar{r}^g - c_g)$	... mixed, equal to the <i>good</i> types that exit
Bad-only, $\phi \leq \underline{\phi}$	$(1 - \pi) \mu_g (\bar{r}^g - c_g)$	... of the <i>good</i> types that exit

Firms, $\Delta S_i \equiv S_i^{q=1} - S_i^{q=0.5}$:

Regime at $q = 0.5$	ΔS_g	ΔS_b
Full pooling, $\phi > 0$	$\mu_g \frac{\pi}{2} \mu_b (c_b - c_g) > 0$	$-\mu_b \left[(1 - \pi)(\bar{r}^b - \bar{r}^g) + \frac{\pi}{2} \mu_g (c_b - c_g) \right] < 0$
Hybrid, $\underline{\phi} < \phi \leq 0$	no closed form, > 0	no closed form, ≤ 0
Bad-only, $\phi \leq \underline{\phi}$	$\mu_g \frac{\pi}{2} (\bar{r}^g - c_g) > 0$	0

Good types gain in every regime, bad types never gain. Under full pooling information only redistributes: $\Delta\Pi + \Delta S_g + \Delta S_b = 0$.

Calibration

Calibration (SMM)

Objective: recover parameters of competition π and signal precision q

Moments matched : Mean: $\mathbb{E}[r]$, Variance: $\text{Var}(r)$, Correlation with default probability: $\text{Corr}(r, p)$

Contribution of information:

$$\frac{\text{Var}(\hat{q}) - \text{Var}(q = 0.5)}{\text{Var}(q = 0.5)} \times 100\%$$

» Details

» Back

Results: Search Frictions vs. Screening

Table 14: Calibrated parameters by rating class

Rating	$\hat{\pi}$	$\hat{q}$	$\hat{r}_g$	Info Contr. (%)	Obj. Q
3+	0.772	0.997	3.461	4.89	0.41
3	0.759	1.000	3.706	4.56	0.35
4+	0.738	0.999	4.609	3.94	0.15
4	0.757	1.000	5.056	5.06	0.19
5+	0.749	1.000	5.403	3.05	0.20
5	0.729	0.944	6.662	0.04	0.04
6	0.721	0.526	7.057	0.00	0.04
7	0.711	0.526	9.350	0.00	0.00

Key findings:

1. **Competitive market:** $\hat{\pi} \in [0.71, 0.77]$ - high search intensity across all classes.
2. **Screening gap:** Safe ratings (3+–5+) near-perfect screening ($\hat{q}$ close to 1); risky ratings (6–7) essentially none ($\hat{q} \approx 0.53$).
3. **Search dominates:** Information contributes at most 5.1% of residual variance.

Calibration: Parameters & Moments

Parameters recovered (3):

- π : search friction (share of shoppers visiting both banks)
- q : signal precision (bank's screening quality)
- $\bar{r}_g$: reservation rate for good types

Moments matched (3):

- Mean: $\mathbb{E}[r]$
- Variance: $\text{Var}(r)$
- Correlation with default probability: $\text{Corr}(r, p)$

Calibrated from data (by rating class):

- Success probabilities p_g, p_b and type shares μ_b from rating transitions
- Upper bound bad types $\bar{r}_b$ from observed $\max(r)$
- Fixed: $r_{CB} = 1$ (average policy rate over the sample period)

Calibration: Strategy

Objective:

minimize squared proportional errors across three moments:

$$Q(\theta) = \sum_m \left(\frac{m^{\text{model}} - m^{\text{data}}}{m^{\text{data}}} \right)^2$$

Numerical implementation:

Grid search (20^3 combinations over $\pi, q, \bar{r}_g$) + L-BFGS-B local refinement

Decomposition (information contribution):

$$\frac{\text{Var}(\hat{q}) - \text{Var}(q = 0.5)}{\text{Var}(q = 0.5)} \times 100\%$$

Back to the data

How to rationalize the differences in different markets?

► dispersion over PD

► high vs low

Low PD firms: low dispersion, high correlation.

- Narrow rate region (parameters)
- High information
- Low/high competition?

High PD firms: high dispersion, no correlation

- Wider rate region (parameters, ex. $R^b \gg R^g$, $p^b \ll p^g$)
- Low information (or cost of acquiring is too high) → pooling is more likely
- Low/high competition?

→ Calibration results are consistent with these patterns (see Calibration section).

Calibration: Data Moments by Rating Class

Parameters and moments taken from data, by rating class

Rating	Mean	p99 ($\bar{r}_b$)	Var.	Corr.	p_g	p_b	N_g	N_b
3+	1.67	3.532	0.406	0.0251	0.0002	0.0032	1 748	9 569
3	1.83	3.785	0.443	0.0326	0.0005	0.0035	5 567	16 620
4+	2.40	4.703	0.553	0.0315	0.0013	0.0078	10 056	28 499
4	2.37	5.199	0.730	0.0478	0.0032	0.0193	18 266	37 204
5+	2.57	5.527	0.867	0.0478	0.0078	0.0347	24 033	23 178
5	3.36	6.798	0.985	0.0174	0.0193	0.0924	15 869	5 796
6	3.63	7.201	1.137	0	0.0193	0.2158	4 641	329
7	4.79	9.541	1.843	0	0.0347	0.3698	1 049	78

Notes: $\bar{r}_b$ (upper bound for bad types) set at the 99th percentile of the observed rate distribution in each class, an outlier-robust analogue of the literal maximum. p_g , p_b : average 3-year default probabilities. N_g , N_b : number of observations per type. Mean, variance, and correlation are the three moments matched in calibration.

Correlation by type

Imagine that within each rating class there are good and bad firms, that you can recognize by looking at the evolution of their default probability.

1. good if PD increases after 10 years, bad if PD decreases, null otherwise
2. Collect residuals and study them separately in each sample

Good rating market:

- good (7th) and bad (28) types .

Mean: $mean^{good} = -0.04 < mean^{bad} = 0.02$. SD=
 $sd^{good} = 0.65 < sd^{bad} = 0.78$, correlation is positive

Bad rating market:

- good (6th) and bad (13) types .

Mean: $mean^{good} = -0.057 < mean^{bad} = 0.05$. SD=
 $sd^{good} = 0.98 < sd^{bad} = 1.19$, correlation is positive

Table 15: Observed vs. model-implied moments

Rating	Mean		Variance		Correlation	
	Data	Model	Data	Model	Data	Model
3+	1.67	1.94	0.406	0.154	0.025	0.025
3	1.83	2.08	0.443	0.188	0.033	0.033
4+	2.40	2.53	0.553	0.338	0.032	0.032
4	2.37	2.63	0.730	0.424	0.048	0.048
5+	2.57	2.80	0.867	0.492	0.048	0.048
5	3.36	3.44	0.985	0.794	0.017	0.017
6	3.63	3.67	1.137	0.913	0.000	0.000
7	4.79	4.78	1.843	1.725	0.000	0.000

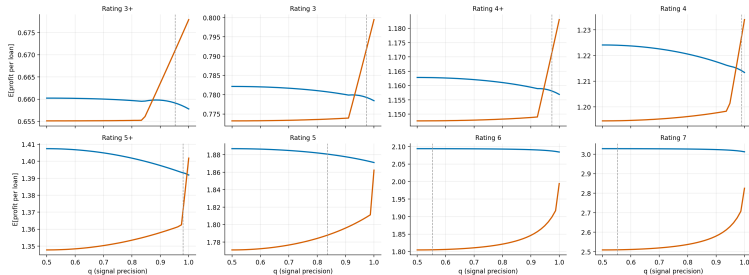
Notes: Correlation is matched essentially exactly in every class, and the mean to within roughly 15%. Variance is matched only partially (38–94% of the data value), consistent with a two-type model confronting a continuum of within-class borrower heterogeneity: it should be read as a lower bound on the dispersion the search-and-screening channel alone can generate.

Surplus sharing and stability

Bank profit by borrower type

Expected bank profit per loan by borrower type vs. signal precision q
(accepted/paid distribution $g(r)$; other parameters fixed at GMM-estimated values; dashed line = estimated $\hat{q}$)

— Profit | good type — Profit | bad type



Expected bank profit per loan, $E[r \mid \text{type}] - c_{\text{type}}$, by true borrower type and rating class, over $q \in [0.5, 1]$. Dashed line: the class's own $\hat{q}$.

Redistribution, as % of each party's own surplus

Change in surplus from $q = 0.5$ to $q = 1$, % of that party's own surplus at $q = 0.5$

Rating	Bad types	Banks	Good types
3+	-1.65	2.43	0.10
3	-1.80	2.19	0.08
4+	-1.82	1.93	0.13
4	-2.27	2.37	0.27
5+	-2.13	1.39	0.31
5	-2.76	0.65	0.39
6	-5.58	0.16	0.28
7	-7.75	0.16	0.46

Rows do not sum. This is the denominator relevant to a bank weighing the cost of precision, and to a borrower judging the size of the repricing.

Is the calibrated $\hat{q}$ an equilibrium?

The calibration holds **both** banks at the same precision. Does either want to deviate?

Hold the rival at $(q^*, F(\cdot; q^*))$ and let one bank pick q_1 and best-respond in price. Changing q_1 moves three objects: the signal shares σ_g, σ_b , the posteriors α^{s_i} , and the pooling cost $c^{avg, s_i} = \alpha^{s_i} c_g + (1 - \alpha^{s_i}) c_b$.

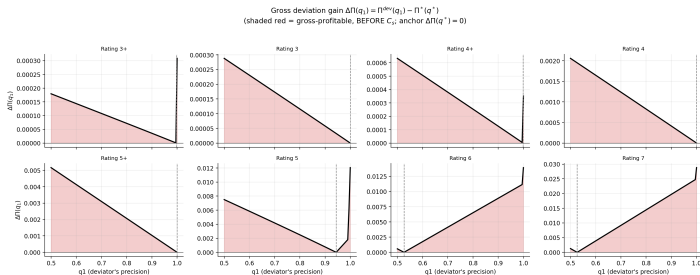
Banks at the same precision agree on every firm; a deviation changes the classification only at the margin. On the share $\min(\sigma_i(q_1), \sigma_i(q^*))$ the two agree and the deviator faces F_i ; on the extra share $\max(\sigma_i(q_1) - \sigma_i(q^*), 0)$ they **disagree** and it faces F_{-i} .

q^* survives a unilateral information deviation iff no Δq satisfies

$$\Delta \Pi(\Delta q) - \Delta C_s(\Delta q) > 0.$$

The best response is a **pure** strategy: once $c^{avg, s_i}(q_1)$ no longer matches the one F_i was built for, profit is no longer flat in r . Flatness survives only at $q_1 = q^*$.

Deviation gains are profitable but small

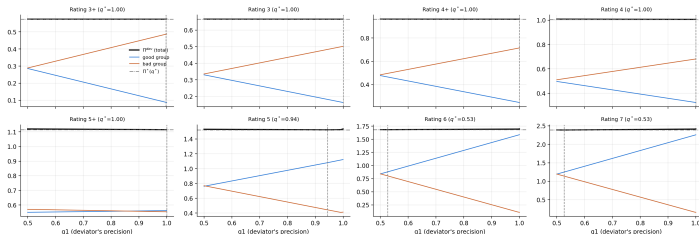


Gain $\Delta\P(q_1) = \Pi^{dev}(q_1) - \Pi^*(q^*)$, by rating class, zero at $q_1 = q^*$ by construction. Shaded: gross-profitable deviations, **before** netting the cost of screening.

$\Delta\P$ rises in both directions but stays on the order of 1% of per-encounter profit. Downward deviations are profitable outright (lower gross profit foregone *and* cost savings); upward ones are ruled out whenever the marginal cost of precision exceeds the gross gain.

Deviation profit, levels

Deviation profit $\Pi^{dev}(q_1)$ -- classification-share coupling
(dash-dot = symmetric $\Pi^*(q^*)$; dashed = q^*)



Gross deviation profit $\Pi^{dev}(q_1)$ against the symmetric benchmark $\Pi^*(q^*)$ (dash-dot), by rating class.